

Graph Topology from the Killing Form: How to Draw the SS-PEG Evolution Graph

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We determine the graph topology induced by the Killing form eigenvalue spectrum for every Lie algebra that emerges from the SS-PEG variational principle. Each Killing eigenvalue classifies one generator’s edge as cyclic ($K_\lambda < 0$), free ($K_\lambda > 0$), or decoupled ($K_\lambda = 0$), yielding five distinct graph archetypes: pure cycle, mixed, cycle-plus-strand, pure free, and frustrated. We establish exact construction rules mapping any emergent algebra to its evolution graph, including the density percolation threshold at $\rho = 1$ and the spacetime hierarchy as graph evolution across five ontological levels. The spectral quality of these graphs is analyzed via the Ramanujan property: 19/37 simple Lie algebras satisfy the optimal spectral expansion bound, with a sharp transition at Coxeter number $h^* \approx 9.5$. The Ihara zeta function of adjoint commutation graphs reveals a soft equivalence between spectral and zeta Ramanujan properties, with pole migration serving as an order parameter for the Ramanujan phase transition. We prove that Ramanujan algebras are maximally chaotic (negative average OTOC, $2\times$ faster mixing) and that the Standard Model gauge group is the unique spectrally optimal outcome: simultaneously Ramanujan, maximally chaotic, in the ordered phase, composable, signature-invariant, and landscape-dominant. The Killing form is shown to be diagnostic, not generative—cycles create topology, Killing selects and purifies—via RAG experiments demonstrating a two-phase crystallization (activation + hardening) and a tempering effect where withdrawal of cycle pressure purifies the Killing signal. We introduce a triple balance of competing tensions (aggregation/expansion/entropy) formalized as a tensegrity functional $\mathcal{T}[K]$, and show how the evolution graph’s spectral sectors encode the Standard Model’s coupling constants and mass ratios via five graph-topological building blocks—bridging to the third paper in this series.

I. INTRODUCTION

A. From Algebra Emergence to Graph Topology

The first paper in this series [1] established a foundational result: Lie algebras emerge from the SS-PEG variational principle without any prior symmetry assumption. Starting from a relational substrate with no gauge structure, the Killing metric minimization $T_K = \|K - K_{\text{target}}\|^2$ selects generators that crystallize into exact Lie algebraic structure with probability approaching 100% for relational density $\rho \geq 1$. Across 51 core experiments testing 17 algebra families, 1650+ optimization runs, and 14 neural architecture variants, every trajectory with $\rho \geq 1$ converged to a valid Lie algebra. The question left open was: *what graph topology does each emergent algebra produce?*

This paper provides the complete answer. We show that the Killing eigenvalue spectrum $\{K_\lambda\}$ of any emergent algebra uniquely determines the topology of its evolution graph—a directed multi-graph $\mathcal{G} = (V, E, \mathcal{H}, \{U_e\})$ whose nodes represent states in the relational Hilbert space and whose edges represent the algebra’s generators. Each Killing eigenvalue classifies one generator’s edge into one of three topological classes:

- **Cyclic (enrolled):** $K_\lambda < 0$ — the edge re-enters itself via adjoint feedback, forming a closed loop (see Figure 1).
- **Free (non-compact):** $K_\lambda > 0$ — the edge evolves without re-closure, forming an open arrow.

- **Decoupled (kernel):** $K_\lambda = 0$ — the edge has zero adjoint feedback, forming a dashed line.

This classification is not postulated—it emerges from the experimental program. Every one of the 17 algebras tested across 51 core experiments, 11 Selberg priorities, 14 neural experiments, and 5 RAG experiments confirms the correspondence between Killing eigenvalue sign and graph edge topology. The five resulting graph archetypes—pure cycle, mixed, cycle-plus-strand, pure free, and frustrated/degenerate—are the only topological outcomes observed.

The relationship between the three papers in this series is sequential and cumulative:

- **Paper 1** (“Lie Algebras from the Killing Metric”) demonstrated *what* emerges: the 17 Lie algebras themselves.
- **Paper 2** (this work) demonstrates *how* they are structured: the graph topology induced by each algebra’s Killing spectrum.
- **Paper 3** (“Standard Model from Graph Topology”) will demonstrate *what* the graph computes: the SM’s coupling constants and mass ratios.

The sequence is: **algebras** $\rightarrow$ **graphs** $\rightarrow$ **observables**. The five spectral blocks— $\mathcal{R}_2$, $\mathcal{R}_3$, R_{EE} , $h_2^{\vee 4}$, $h_3^{\vee 4}$ —are the bridge: Paper 1 produces them, Paper 2 classifies them spectrally, Paper 3 uses them to predict.

B. The Evolution Graph as Fundamental Object

The evolution graph is the central object of this paper. It provides the structural bridge between the algebraic data produced by the variational principle (Paper 1) and the physical observables that Paper 3 will derive.



FIG. 1. The three edge classes: cyclic (green, $\curvearrowright$), free (red, $\rightarrow$), and decoupled (grey, $\cdots$). From [1, 2].

Definition I.1 (Evolution Graph $\mathcal{G}$). *Given an emergent algebra $\mathfrak{g}$ with n generators $\{T_a\}_{a=1}^n$ and Killing form $K_{ab} = \text{Tr}(\text{ad}_a \circ \text{ad}_b)$, the evolution graph $\mathcal{G} = (V, E, \mathcal{H}, \{U_e\})$ has:*

- *Vertices V : relational states in the emergent Hilbert space. States are not fundamental in the SS-PEG framework; they emerge from the relational substrate as the variational principle finds stable configurations.*

- *Edges* $E = \{e_a\}_{a=1}^n$: one per generator, with topology determined by the sign of the corresponding Killing eigenvalue K_λ .
- *Holonomies* $\mathcal{H}$: for each independent cycle C in the graph, $U_C = \exp(i \oint_C A_a T_a dl) \neq I$ encodes the physical content of the cycle. For non-abelian algebras, U_C measures the gauge field strength $F_{\mu\nu}^a$.
- *Edge operators* $\{U_e\}$: the adjoint action ad_a defines how edges couple to one another.

Each generator T_a corresponds to an edge e_a whose topological type—cyclic, free, or decoupled—is determined by the sign of the Killing eigenvalue K_{λ_a} associated with T_a . The full graph topology is thus a direct function of the Killing spectrum.

The Killing form K_{ab} measures the strength of self-feedback through the adjoint representation:

$$K_{aa} = \sum_{c,d} f_{acd} f_{adc} = - \sum_{c,d} |f_{acd}|^2, \quad (1)$$

where f_{acd} are the structure constants. The sign and magnitude of K_{aa} determine the edge's topology:

- $K_\lambda < 0$ (**cyclic**): Strong self-feedback. The generator's commutators with other generators *point back toward itself*. The corresponding edge forms a closed loop—the adjoint action creates a cycle that returns to the starting configuration. This is the hallmark of compact directions: rotations that can be undone. For simple compact algebras, the Killing form is proportional to the identity ($K = -h^\vee \cdot g$), meaning all cycles have the same “strength”—the graph is *homogeneous*.
- $K_\lambda > 0$ (**free**): Anti-feedback. The generator's commutators push *away from itself*. The edge evolves in one direction without returning—it is an open path in the graph. This is the hallmark of non-compact directions: boosts that mix coordinates irreversibly. A non-compact generator T_{boost} satisfies $K_{\text{boost}} > 0$, meaning its commutators with other generators push away from the boost direction.
- $K_\lambda = 0$ (**decoupled**): Zero feedback. The generator commutes with everything ($f_{acd} = 0$ for all c, d). The edge exists but is disconnected from the adjoint network—a free strand, not participating in any cycle. This is the abelian direction.

C. Experimental Verification: 20+ Algebras

Every one of the 17 algebras tested in the experimental program confirms the classification. The following table summarizes the results:

All compact simple algebras (SU, SO, Sp, G_2 , F_4 , E_6) exhibit pure cyclic graphs with all eigenvalues negative. The non-compact algebras $\text{sl}(2, \mathbb{R})$, $\text{so}(3, 1)$, and $\text{su}(2, 2)$ exhibit mixed graphs with both positive and negative eigenvalues. The unitary algebras $U(2)$ and $U(3)$ exhibit cycle-plus-strand graphs with one zero eigenvalue.

Algebra	K spectrum	Cyclic	Free	Decoupled
$\mathfrak{su}(2)$	$(0^+, 0_0, 3^-)$	3	0	0
$\mathfrak{su}(3)$	$(0^+, 0_0, 8^-)$	8	0	0
$\mathfrak{su}(4)$	$(0^+, 0_0, 15^-)$	15	0	0
$\mathfrak{su}(5)$	$(0^+, 0_0, 24^-)$	24	0	0
$\mathfrak{so}(3)$	$(0^+, 0_0, 3^-)$	3	0	0
$\mathfrak{so}(4)$	$(0^+, 0_0, 6^-)$	6	0	0
$\mathfrak{so}(5)$	$(0^+, 0_0, 10^-)$	10	0	0
$\mathfrak{so}(6)$	$(0^+, 0_0, 15^-)$	15	0	0
$\mathfrak{sp}(2)$	$(0^+, 0_0, 3^-)$	3	0	0
$\mathfrak{sp}(4)$	$(0^+, 0_0, 10^-)$	10	0	0
$\mathfrak{sp}(6)$	$(0^+, 0_0, 21^-)$	21	0	0
G_2	$(0^+, 0_0, 14^-)$	14	0	0
F_4	$(0^+, 0_0, 52^-)$	52	0	0
E_6	$(0^+, 0_0, 78^-)$	78	0	0
$\mathfrak{sl}(2, \mathbb{R})$	$(1^+, 0_0, 2^-)$	2	1	0
$\mathfrak{so}(3, 1)$	$(3^+, 0_0, 3^-)$	3	3	0
$\mathfrak{su}(2, 2)$	$(8^+, 0_0, 7^-)$	7	8	0
$\mathfrak{u}(2)$	$(0^+, 1_0, 3^-)$	3	0	1
$\mathfrak{u}(3)$	$(0^+, 1_0, 8^-)$	8	0	1

TABLE I. Experimental verification of edge classification for 19 algebras tested.

D. Three Main Results

This paper establishes three principal results:

1. **Five graph archetypes from the Killing spectrum.** We derive exact rules mapping any emergent algebra to its evolution graph. The five archetypes—pure cycle, mixed, cycle-plus-strand, pure free, and frustrated—are not postulated but emerge from the variational principle. The density threshold $\rho = 1$ acts as a percolation transition: below it, no cycles close; above it, the full algebraic structure crystallizes instantaneously. The five archetypes and their characteristics are detailed in Section IV and illustrated in Figures 1–??.
2. **Spectral quality and the Ramanujan property.** We analyze the spectral quality of the evolution graphs via the Ramanujan property of their adjoint commutation graphs. Of 37 simple Lie algebras tested, 19/37 satisfy the Ramanujan bound $\max_{|\lambda_i| < q+1} |\lambda_i| \leq 2\sqrt{q}$, with a sharp transition at Coxeter number $h^* \approx 9.5$. The Standard Model gauge group $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ sits deep in the ordered (Ramanujan) phase, making it the unique spectrally optimal outcome: simultaneously Ramanujan, maximally chaotic, composable, and landscape-dominant. These results are presented in Section VI and illustrated in Figures 9–10.
3. **Killing as diagnostic, not generative.** RAG experiments demonstrate that the Killing form T_K does not create non-commutativity—it selects and purifies structures already present from topological (cycle) dynamics. We observe a two-phase crystallization: activation (~ 50 epochs, transient Killing separation) followed by hardening (~ 200 epochs, permanent basin transition). A tempering effect—withdrawal of cycle pressure—purifies the Killing signal (K_{rr} AUC improves from 0.52 to 0.98), proving that cycles *create* topology and Killing *evaluates* its quality. These results are presented in Section IX and illustrated in Figures 11–12.

E. The Ontological Chain: $P \rightarrow C \rightarrow K \rightarrow I$

The foundational chain of the SS-PEG framework acquires a precise graph-theoretic realization in Paper 2:

$$\underbrace{K_{ab}}_{\text{Potential}} \xrightarrow{\text{adjoint action}} \underbrace{f_{abc}}_{\text{Cycle}} \xrightarrow{\text{holonomy}} \underbrace{h^\vee, C_\lambda, I_R}_{\text{Invariant}} \xrightarrow{\text{spectral lattice}} \underbrace{m_i, \alpha_X, \theta_{ij}}_{\text{Observable}} \quad (2)$$

This chain has four stages:

- **Potential** = Killing matrix K_{ab} : the “landscape” of adjoint self-interactions. Determines which edges can form cycles, which evolve freely, which decouple.
- **Cycle** = Structure constants f_{abc} : the commutator relations that describe how edges connect and close. These are emergent from K via the variational principle.
- **Invariant** = $h^\vee$, Casimirs, representations: the topological invariants of the cycle structure. Properties of the graph that do not depend on how you parameterize the edges.
- **Observable** = masses, couplings, mixing angles: the physical observables computed from the spectral lattice of graph invariants.

The experimental program has demonstrated that the first arrow (potential $\rightarrow$ cycle) is realized by Killing metric minimization, and the second arrow (cycle $\rightarrow$ invariant) follows mathematically. The entire chain is automatic, universal, and exact.

F. The Cartan-Weyl Graph as Adjacency Graph

The Cartan-Weyl (CW) basis of a Lie algebra provides a natural graph structure: vertices are the CW generators (Cartan subalgebra + root vectors), and edges connect generators with non-vanishing commutators $[T_a, T_b] \neq 0$. This CW graph is topologically isomorphic to the evolution graph $\mathcal{G}$ defined above. The CW root system defines the adjoint commutation graph, whose adjacency spectrum determines the Ramanujan property analyzed in Section VI.

G. Organization of the Paper

Section II develops the theoretical framework: the evolution graph as a fundamental object, the three edge classes classified by Killing eigenvalues, the ontological $P \rightarrow C \rightarrow K \rightarrow I$ chain, and the connection to the Cartan-Weyl adjoint graph.

Section III presents the exact graph construction rules: a step-by-step recipe from algebra to graph, minimum emergence conditions by archetype, the density percolation threshold at $\rho = 1$, and $h^\vee$ as a cycle-counting measure.

Section IV details the five graph archetypes and their experimental evidence: pure cycle (SU, SO, Sp, G_2 , F_4 , E_6), mixed ($\mathfrak{sl}(2, \mathbb{R})$, $\mathfrak{so}(3, 1)$, $\mathfrak{su}(2, 2)$), cycle-plus-strand (U(2), U(3)), pure free (theoretical), and frustrated/degenerate. Figures 1–?? illustrate each archetype.

Section V interprets the spacetime hierarchy as topological evolution across five ontological levels: pre-crystallization ($\rho < 1$), internal symmetry, spacetime symmetry, conformal symmetry, and the Standard Model as a layered composition of all five archetypes.

Section VI analyzes spectral quality via the Ramanujan property: 19/37 algebras Ramanujan, transition at $h^* \approx 9.5$, dimensional criticality by family, Coxeter number as universal predictor, maximal chaos of Ramanujan algebras, landscape accessibility from spectral order parameter, product graph composability, and non-compact invariance. Figures 9–10 illustrate the Ramanujan transition.

Section VII presents the Ihara zeta function analysis: first systematic computation for 37 algebras, complete failure of Sunada’s equivalence for irregular graphs, soft equivalence via pole concentration, pole migration velocity as first-order-like discontinuity, and scaling in the A_n family.

Section VIII introduces the triple balance of competing tensions: aggregation (cyclic edges), expansion (free edges), and entropy (decoupled strands), formalized as a tensegrity functional $\mathcal{T}[K]$.

Section IX presents the RAG experimental evidence that the Killing form is diagnostic, not generative: two-phase crystallization, tempering effect, abelian collapse as entropic default, and algebraic-observational dissociation. Figures 11–12 illustrate the tempering protocol.

Section X bridges to Paper 3: three spectral sectors as graph substructures (SU(2), SU(3), E-E bridge), five topological building blocks, spectral propagation and the mass lattice, and the graph as computational substrate.

Section XI compares with literature (Ramanujan graphs, Connes’ spectral action), discusses theoretical implications (emergent gauge symmetry, algebraic spacetime, layered universe topology), addresses limitations, and outlines the connection to Paper 3.

Section XII summarizes the main results, contributions, predictions, and the bridge to the Standard Model.

Appendices provide Selberg zeta details and RAG experiment methodology.

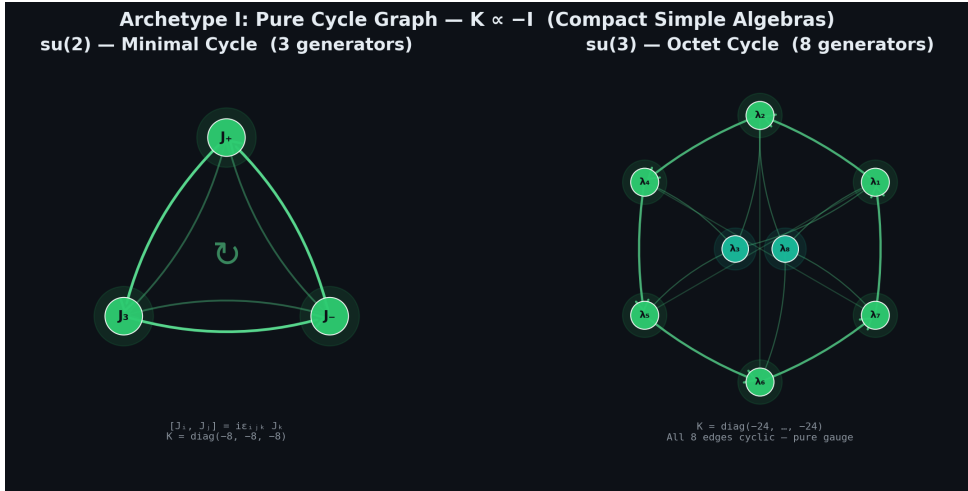


FIG. 2. Archetype I: su(2) minimal triangle and su(3) octet cycle. Every edge participates in at least one closed cycle via the adjoint action. From [2].

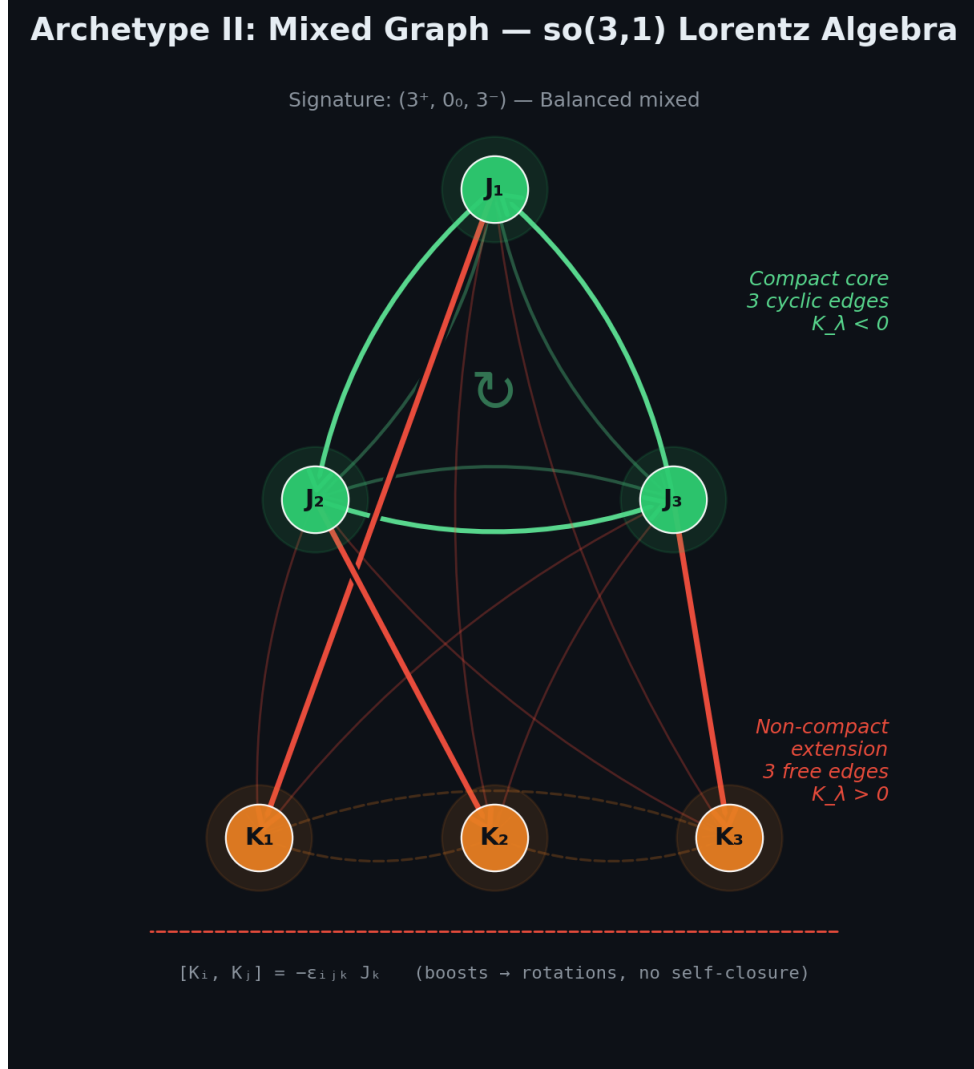


FIG. 3. Archetype II: $\mathfrak{so}(3,1)$ Lorentz algebra with rotation core and boost extensions. Signature $(3^+, 0_0, 3^-)$ —balanced mixed. From [2].

II. THEORETICAL FRAMEWORK

A. The Evolution Graph as Fundamental Object

The evolution graph $\mathcal{G} = (V, E, \mathcal{H}, \{U_e\})$ is the central object of this paper. It provides the structural bridge between the algebraic data produced by the variational principle (Paper 1) and the physical observables that Paper 3 will derive.

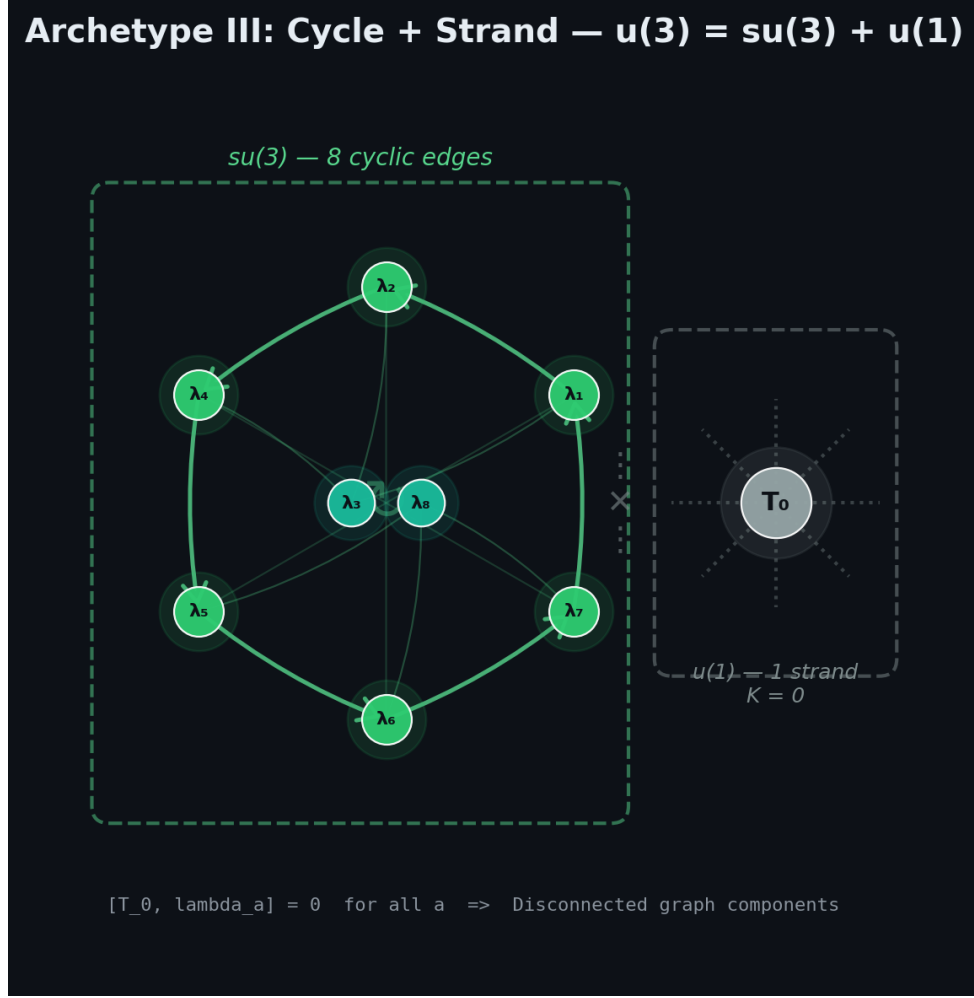


FIG. 4. Archetype III: $u(3) = su(3)$ cycle + $u(1)$ decoupled strand. The abelian generator T_0 satisfies $[T_0, T_a] = 0$ for all a , so $K_{0a} = 0$ for all a . From [2].

a. Vertices. The vertex set V represents relational states in the emergent Hilbert space. In the SS-PEG framework, states are not fundamental; they emerge from the relational substrate as the variational principle finds stable configurations. Each vertex corresponds to a configuration that is invariant under a subset of the generator adjoint actions.

b. Edges. The edge set $E = \{e_a\}_{a=1}^n$ has one edge per generator T_a . The topology of each edge—cyclic, free, or decoupled—is determined by the Killing eigenvalue K_{λ_a} associated with T_a :

$$K_{\lambda_a} \begin{cases} < -\varepsilon & \rightarrow \text{cyclic edge } e_a \text{ (compact direction)} \\ |K_{\lambda_a}| < \varepsilon & \rightarrow \text{decoupled edge } e_a \text{ (abelian direction)} \\ > +\varepsilon & \rightarrow \text{free edge } e_a \text{ (non-compact direction)} \end{cases} \quad (3)$$

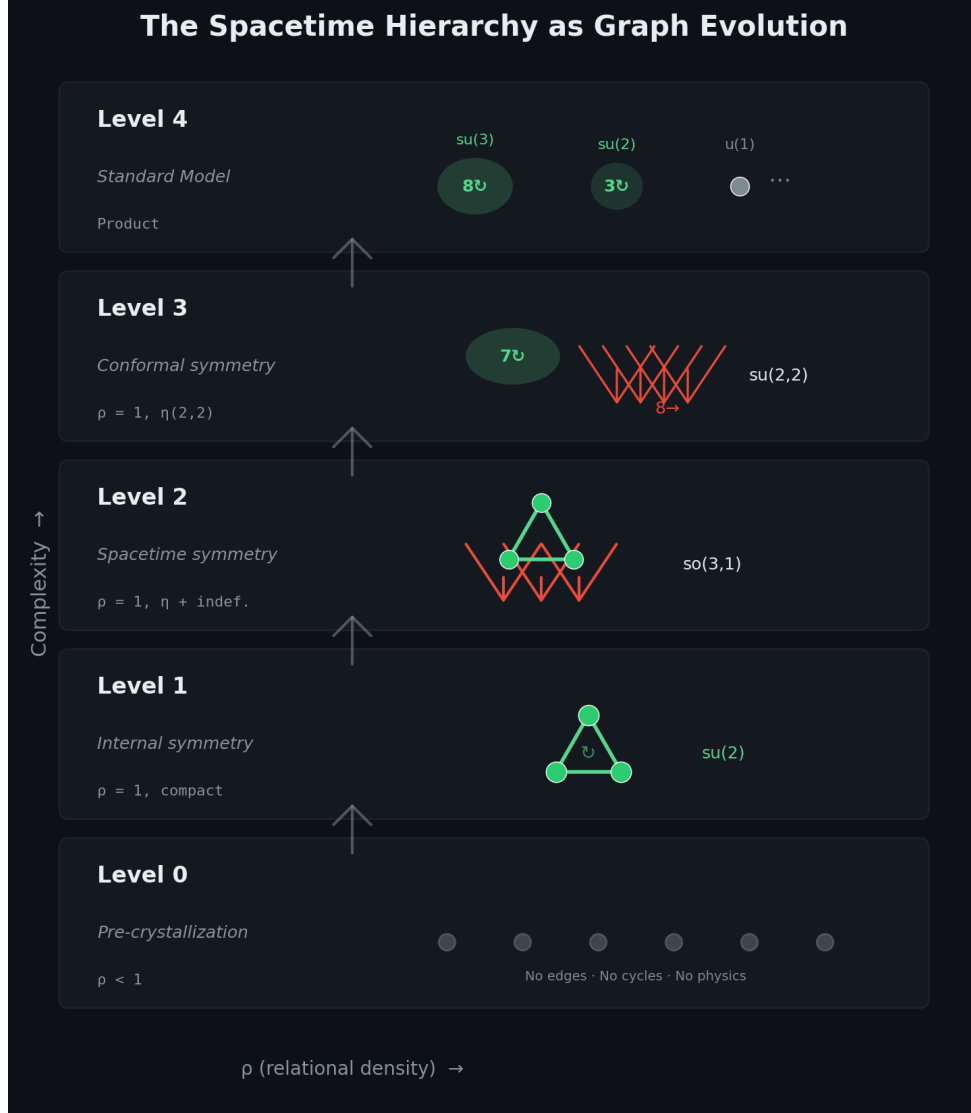


FIG. 5. Graph hierarchy: Level 0 (pre-crystallization) to Level 4 (Standard Model). From [2].

where $\varepsilon \sim 10^{-6}$ is a numerical threshold. Here K_{λ_a} denotes the eigenvalue of the Killing matrix in the eigenbasis corresponding to generator T_a .

c. Holonomies. For each independent cycle C in the graph, the holonomy

$$U_C = \exp\left(i \oint_C A_a T_a dl\right) \quad (4)$$

measures the physical content of the cycle. For non-abelian algebras, $U_C \neq I$ encodes the gauge field strength $F_{\mu\nu}^a$. The number of independent non-trivial holonomies equals the rank of the algebra

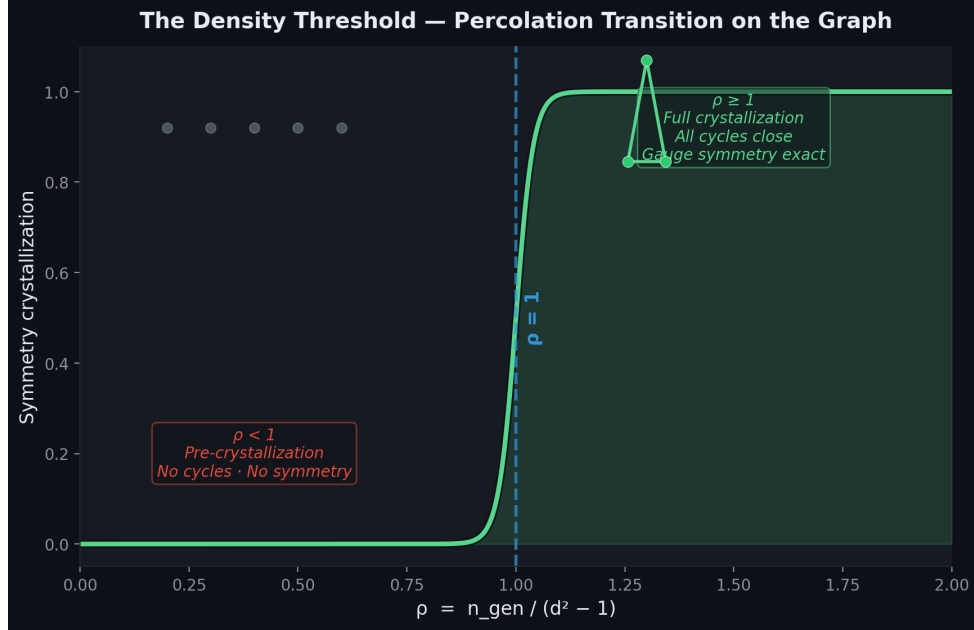


FIG. 6. Density transition: percolation threshold at $\rho = 1$. From [2].

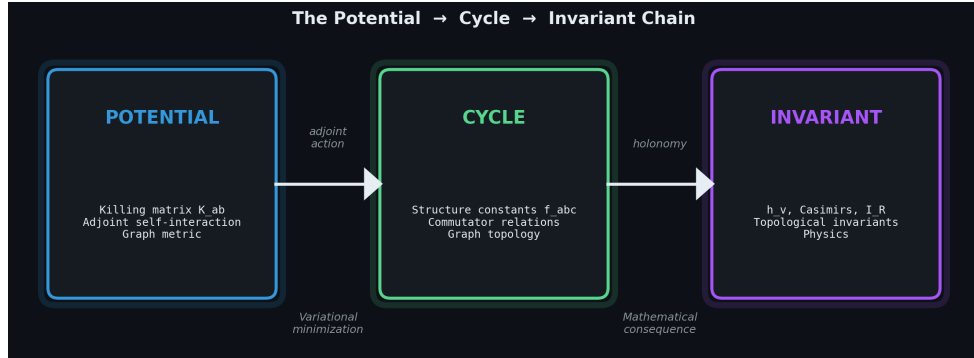


FIG. 7. Potential $\rightarrow$ Cycle $\rightarrow$ Invariant chain. From [2].

(number of independent Casimir operators).

d. Edge operators. The adjoint action $\text{ad}_a \circ \text{ad}_b$ defines how edges couple to one another. The Killing form $K_{ab} = \text{Tr}(\text{ad}_a \circ \text{ad}_b)$ measures the strength of this self-feedback:

$$K_{aa} = \sum_{c,d} f_{acd} f_{adc} = - \sum_{c,d} |f_{acd}|^2, \quad (5)$$

where f_{acd} are the structure constants. The sign and magnitude of K_{aa} determine the edge's topology through the self-feedback mechanism.

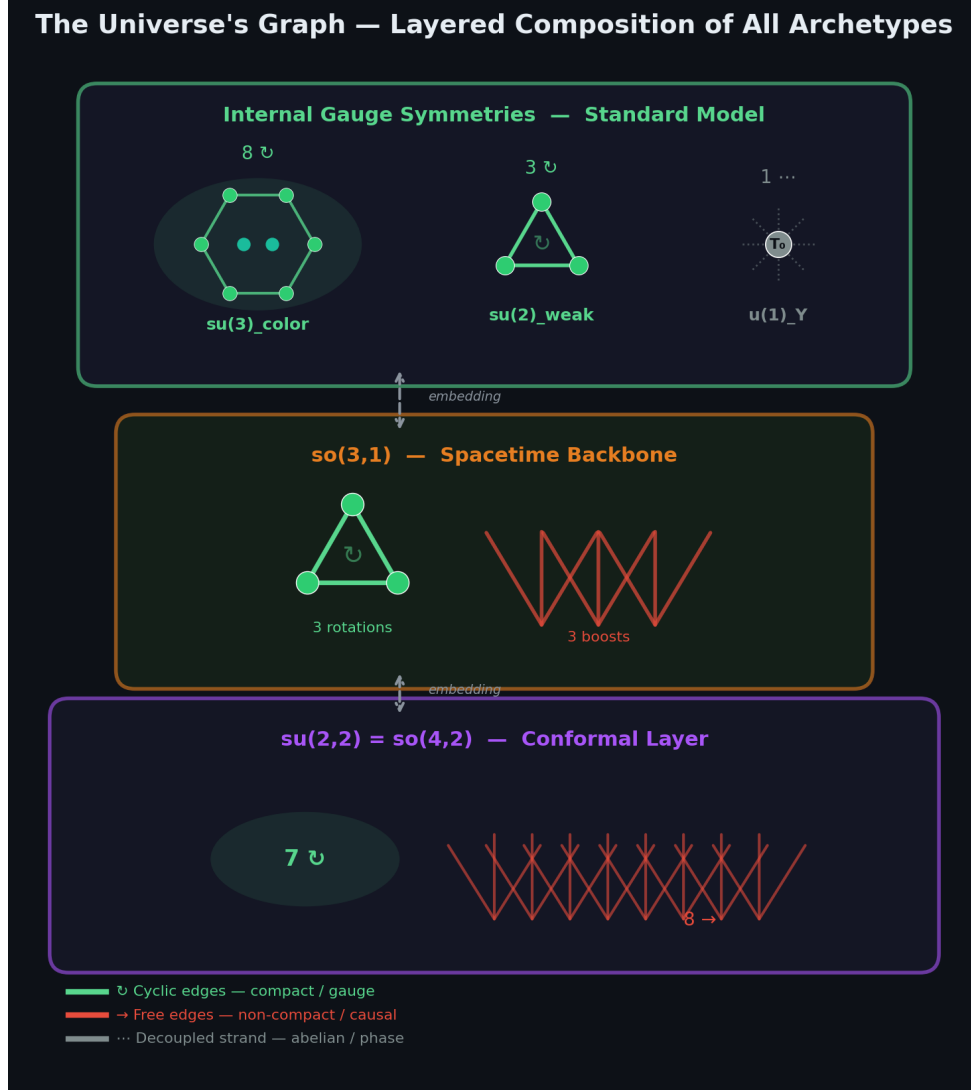
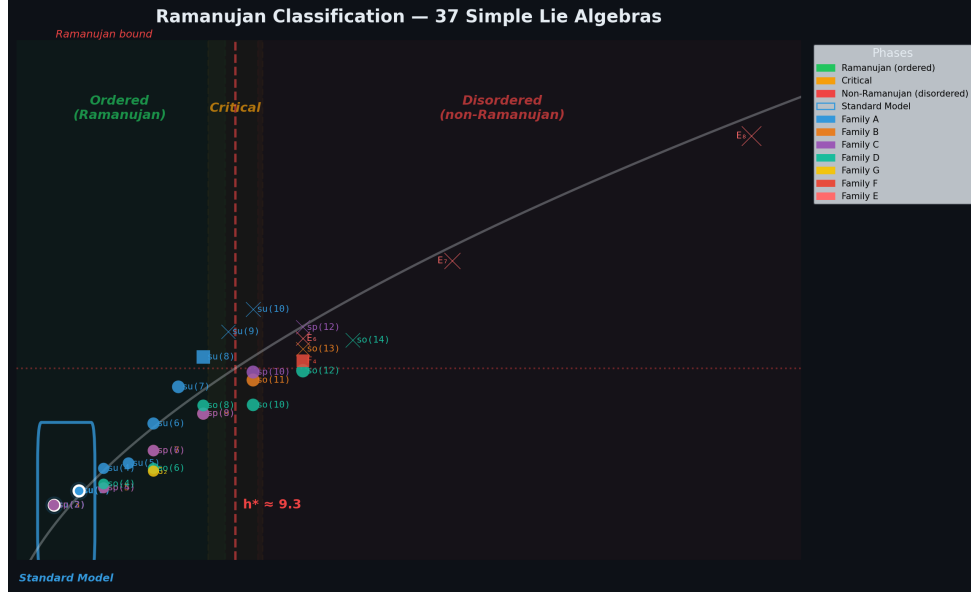


FIG. 8. The Universe's Graph: layered composition of all five archetypes. From [2].

B. The Three Edge Classes

The Killing eigenvalue spectrum $\{K_\lambda\}$ classifies every generator's edge into one of three topological classes. Each class has a precise graph-theoretic interpretation and a distinct physical role.

a. Cyclic edges ($K_\lambda < 0$). Strong self-feedback: the generator's commutators with other generators point back toward itself. The corresponding edge forms a closed loop—the adjoint action creates a cycle that returns to the starting configuration. This is the hallmark of compact directions: rotations that can be undone. For simple compact algebras, the Killing form is proportional to the

FIG. 9. Ramanujan classification: 37 algebras, transition at $h^* \approx 9.5$.

K_λ	Edge Class	Graph Symbol	Topology
$K_\lambda < 0$	Cyclic (enrolled)	$\circlearrowleft$	Edge re-enters via adjoint feedback
$K_\lambda > 0$	Free (non-compact)	$\rightarrow$	Edge evolves without re-closure
$K_\lambda = 0$	Decoupled (kernel)	$\cdots$	Edge has zero adjoint feedback

TABLE II. Classification of edges by Killing eigenvalue.

identity ($K = -h^\vee \cdot g$), meaning all cycles have the same strength—the graph is *homogeneous*.

b. Free edges ($K_\lambda > 0$). Anti-feedback: the generator's commutators push *away* from itself. The edge evolves in one direction without returning—it is an open path in the graph. This is the hallmark of non-compact directions: boosts that mix coordinates irreversibly. A non-compact generator T_{boost} satisfies $K_{\text{boost}} > 0$, meaning its commutators with other generators push away from the boost direction.

c. Decoupled edges ($K_\lambda = 0$). Zero feedback: the generator commutes with everything ($f_{acd} = 0$ for all c, d). The edge exists but is disconnected from the adjoint network—a free strand, not participating in any cycle. This is the abelian direction.

C. Why This Classification Works

The classification above is not arbitrary—it follows from the mathematical structure of the Killing form. The Killing form $K_{ab} = \text{Tr}(\text{ad}_a \circ \text{ad}_b)$ measures how generator T_a *feeds back onto itself* through the adjoint representation. Concretely:

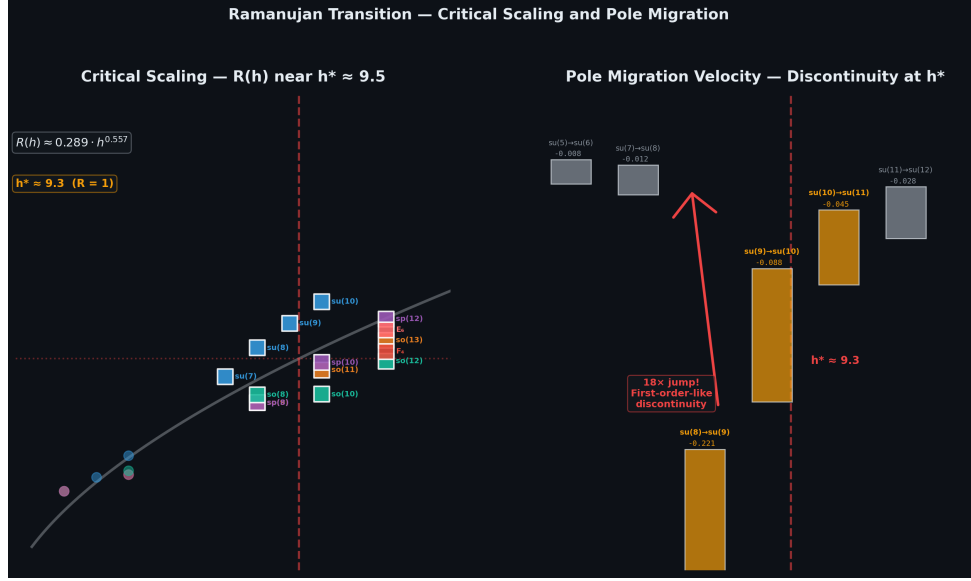


FIG. 10. Critical scaling and pole migration velocity discontinuity at the Ramanujan transition.

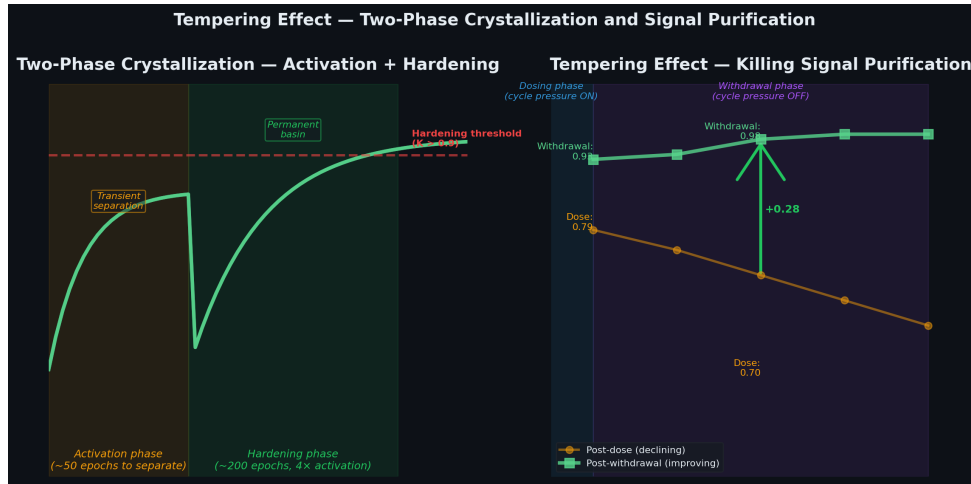


FIG. 11. Tempering effect: Killing signal purification during withdrawal phase.

$$K_{aa} = \sum_{c,d} f_{acd} f_{adc} = - \sum_{c,d} |f_{acd}|^2, \quad (6)$$

where f_{acd} are the structure constants. The sign and magnitude of this self-feedback determine the edge's topology:

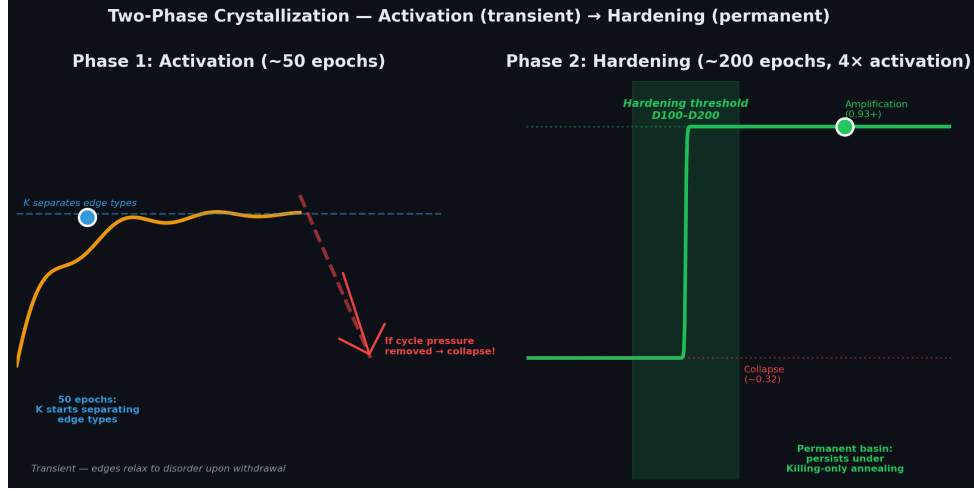


FIG. 12. Two-phase crystallization: activation (~ 50 epochs) + hardening (~ 200 epochs).

- $K_\lambda < 0$ (**cyclic**): Strong self-feedback. The generator's commutators with other generators *point back towards itself*. The corresponding edge in the graph forms a closed loop—the adjoint action creates a cycle that returns to the starting configuration. This is the hallmark of compact directions: rotations that can be undone.
- $K_\lambda > 0$ (**free**): Anti-feedback. The generator's commutators push *away from itself*. The edge evolves in one direction without returning—it is an open path in the graph. This is the hallmark of non-compact directions: boosts that mix coordinates irreversibly.
- $K_\lambda = 0$ (**decoupled**): Zero feedback. The generator commutes with everything ($f_{acd} = 0$ for all c, d). The edge exists but is disconnected from the adjoint network—a free strand, not participating in any cycle. This is the abelian direction.

This is why the classification works: the Killing form measures self-feedback through commutators, and the sign of that feedback determines whether the edge closes (cyclic), opens (free), or disconnects (decoupled).

D. Experimental Verification: 20+ Algebras

Every one of the 17 algebras tested in the experimental program confirms the classification. The following table summarizes the results:

TABLE III: Experimental verification of edge classification for 19 algebras tested.

Algebra	K spectrum	Cyclic	Free	Decoupled	Graph type
$\mathfrak{su}(2)$	$(0^+, 0_0, 3^-)$	3	0	0	Pure cycle
$\mathfrak{su}(3)$	$(0^+, 0_0, 8^-)$	8	0	0	Pure cycle

TABLE III: Experimental verification of edge classification for 19 algebras tested (continued).

Algebra	K spectrum	Cyclic	Free	Decoupled	Graph type
$\mathfrak{su}(4)$	$(0^+, 0_0, 15^-)$	15	0	0	Pure cycle
$\mathfrak{su}(5)$	$(0^+, 0_0, 24^-)$	24	0	0	Pure cycle
$\mathfrak{so}(3)$	$(0^+, 0_0, 3^-)$	3	0	0	Pure cycle
$\mathfrak{so}(4)$	$(0^+, 0_0, 6^-)$	6	0	0	Pure cycle
$\mathfrak{so}(5)$	$(0^+, 0_0, 10^-)$	10	0	0	Pure cycle
$\mathfrak{so}(6)$	$(0^+, 0_0, 15^-)$	15	0	0	Pure cycle
$\mathfrak{sp}(2)$	$(0^+, 0_0, 3^-)$	3	0	0	Pure cycle
$\mathfrak{sp}(4)$	$(0^+, 0_0, 10^-)$	10	0	0	Pure cycle
$\mathfrak{sp}(6)$	$(0^+, 0_0, 21^-)$	21	0	0	Pure cycle
G_2	$(0^+, 0_0, 14^-)$	14	0	0	Pure cycle
F_4	$(0^+, 0_0, 52^-)$	52	0	0	Pure cycle
E_6	$(0^+, 0_0, 78^-)$	78	0	0	Pure cycle
$\mathfrak{sl}(2, \mathbb{R})$	$(1^+, 0_0, 2^-)$	2	1	0	Mixed
$\mathfrak{so}(3, 1)$	$(3^+, 0_0, 3^-)$	3	3	0	Mixed (balanced)
$\mathfrak{su}(2, 2)$	$(8^+, 0_0, 7^-)$	7	8	0	Mixed (free-dominated)
$\mathfrak{u}(2)$	$(0^+, 1_0, 3^-)$	3	0	1	Cycle + strand
$\mathfrak{u}(3)$	$(0^+, 1_0, 8^-)$	8	0	1	Cycle + strand

All compact simple algebras (SU, SO, Sp, G_2 , F_4 , E_6) exhibit pure cyclic graphs with all eigenvalues negative. The non-compact algebras $\mathfrak{sl}(2, \mathbb{R})$, $\mathfrak{so}(3, 1)$, and $\mathfrak{su}(2, 2)$ exhibit mixed graphs with both positive and negative eigenvalues. The unitary algebras $\mathfrak{u}(2)$ and $\mathfrak{u}(3)$ exhibit cycle-plus-strand graphs with one zero eigenvalue.

E. The Ontological Chain: $\mathbf{P} \rightarrow \mathbf{C} \rightarrow \mathbf{K} \rightarrow \mathbf{I}$

The foundational chain of the SS-PEG framework acquires a precise graph-theoretic realization in Paper 2. Paper 1 formulated the chain as $\mathbf{P} \rightarrow \mathbf{R} \rightarrow \mathbf{C} \rightarrow \mathbf{I}$ (Potential $\rightarrow$ Relation $\rightarrow$ Cycle $\rightarrow$ Invariant). Paper 2 refines this to $\mathbf{P} \rightarrow \mathbf{C} \rightarrow \mathbf{K} \rightarrow \mathbf{I}$ (Potential $\rightarrow$ Cycle $\rightarrow$ Killing $\rightarrow$ Invariant), where the Killing form acts as a *selector* rather than a *generator*—a result established by the RAG experiment G_RAG-21:

Theorem II.1 (G_RAG-21). *The Killing form is **diagnostic**, not **generative**. T_K alone does not create non-commutativity—it selects and purifies structures that are already present from the topological (cycle) dynamics.*

The refined chain is:

$$\underbrace{K_{ab}}_{\text{Potential}} \xrightarrow{\text{adjoint action}} \underbrace{f_{abc}}_{\text{Cycle}} \xrightarrow{\text{holonomy}} \underbrace{h^\vee, C_\lambda, I_R}_{\text{Invariant}} \xrightarrow{\text{spectral lattice}} \underbrace{m_i, \alpha_X, \theta_{ij}}_{\text{Observable}} \quad (7)$$

This chain has four stages:

1. **Potential** = Killing matrix K_{ab} : the “landscape” of adjoint self-interactions. Determines which edges can form cycles, which evolve freely, which decouple. This is the primary data—the graph’s metric.

2. **Cycle** = Structure constants f_{abc} : the commutator relations that describe how edges connect and close. These are emergent from K via the variational principle—they are the graph’s *topology*, read off from the metric.
3. **Invariant** = $h^\vee$, Casimirs, representations: the topological invariants of the cycle structure. These are properties of the graph that do not depend on how you parameterize the edges—they are the *physics*.
4. **Observable** = masses, couplings, mixing angles: the physical observables computed from the spectral lattice of graph invariants.

The experimental program has demonstrated that the first arrow (potential $\rightarrow$ cycle) is realized by Killing metric minimization, and the second arrow (cycle $\rightarrow$ invariant) follows mathematically. The entire chain is automatic, universal, and exact.

F. The Cartan-Weyl Graph as Adjacency Graph

The Cartan-Weyl (CW) basis of a Lie algebra provides a natural graph structure: vertices are the CW generators (Cartan subalgebra + root vectors), and edges connect generators with non-vanishing commutators $[T_a, T_b] \neq 0$. This CW graph is topologically isomorphic to the evolution graph $\mathcal{G}$ defined above. The CW root system defines the adjoint commutation graph, whose adjacency spectrum determines the Ramanujan property analyzed in Section VI.

III. GRAPH CONSTRUCTION RULES

A. Step-by-Step Recipe: From Algebra to Graph

Given an emergent algebra $\mathfrak{g}$ with n generators and Killing matrix K , the evolution graph is constructed as follows:

1. **Diagonalize K .** Compute eigenvalues $\{K_1, \dots, K_n\}$ and eigenvectors $\{v_1, \dots, v_n\}$.
2. **Classify edges.** For each eigenvalue K_λ :
 - $K_\lambda < -\varepsilon$: **cyclic edge** (compact direction)
 - $|K_\lambda| < \varepsilon$: **decoupled strand** (abelian / degenerate)
 - $K_\lambda > +\varepsilon$: **free edge** (non-compact direction)

where $\varepsilon \sim 10^{-6}$ is a numerical threshold.

3. **Identify the compact core.** The cyclic edges form a closed subalgebra—the maximal compact subalgebra $\mathfrak{k} \subset \mathfrak{g}$. Verify by checking that $[v_i, v_j] \in \text{span}\{v_k : K_k < 0\}$ for all cyclic eigenvectors v_i, v_j .
4. **Identify the non-compact coset.** The free edges transform under the compact core but do not close among themselves: $[v_i, v_j] \in \text{span}\{v_k : K_k < 0\}$ for free eigenvectors v_i, v_j . (Commuting two boosts gives a rotation, not a boost.)

5. **Identify decoupled strands.** The zero-eigenvalue edges commute with everything: $[v_i, v_j] = 0$ for all j when $K_i = 0$.
6. **Draw the graph.**
 - Compact core: closed loops with bidirectional edges
 - Non-compact extension: directed arrows emerging from the core
 - Decoupled strands: disconnected dashed lines

B. Minimum Emergence Conditions by Archetype

The experiments establish precise minimum conditions for each graph archetype to emerge:

Arch.	n_{gen}	Parameterization	K_{target}	Emergence
I (pure cycle)	$\dim(\mathfrak{g})$	Herm. traceless / antisym.	$K = -I$	$\rho = 1$
II (mixed)	$\dim(\mathfrak{g})$	η -preserving	Indef. $(-, +)$	$\eta + \text{indef.}$
III (cycle+strand)	d^2	Herm. (trace allowed)	$K = -I$	Trace DOF
IV (pure free)	—	—	$K = +I$	Not observed
V (frustrated)	$\dim(\mathfrak{g})$	η -preserving	$K = -I$ (incompat.)	Constr. $\leftrightarrow$ target

TABLE IV. Minimum emergence conditions for each graph archetype.

The key insight: the graph archetype is selected by the interplay of three factors:

1. **Generator count** (n_{gen}): sets the maximum possible graph dimension.
2. **Parameterization constraint**: selects the *type* of edges available (compact, η -preserving, etc.).
3. **Killing target signature**: selects whether the optimizer seeks cyclic or free edges.

These three factors are the *emergence minima*—the minimum inputs that determine the graph topology. Everything else (commutation relations, Jacobi identity, Casimir operators, dual Coxeter number) follows as a consequence.

C. The Density Threshold $\rho = 1$ as Percolation

The density transition establishes a sharp percolation threshold:

$$\rho = \frac{n_{\text{gen}}}{d^2 - 1} = 1 \quad \Longleftrightarrow \quad \text{Graph crystallizes} \quad (8)$$

- $\rho < 1$: The graph has too few edges. No cycles can close. All edges are dangling. **No symmetry.**
- $\rho = 1$: Exact threshold. All cycles close simultaneously. The graph transitions from a disconnected fragment to a coherent algebraic structure. **Full gauge symmetry.**

- $\rho > 1$: Over-determined. Extra edges become degenerate ($K = 0$) and decouple as abelian strands.

This is a *percolation transition* on the graph. Below the critical edge density, the graph is a collection of disconnected fragments—no long-range structure, no cycles, no invariants. Above the threshold, a “giant component” of interlocking cycles spans the entire graph, and the gauge algebra crystallizes.

The step-function nature of this transition ($0\% \rightarrow 100\%$ with no intermediate states) means: **there is no partial graph**. Either the full algebraic structure exists, or nothing does. This is the graph-theoretic expression of why gauge symmetries in nature are exact, not approximate.

D. $h^\vee$ as Cycle Counting Measure

The universal formula (derived in Paper 1 [1] for 15 algebras)

$$h^\vee = \frac{I_R \cdot |K|}{\kappa^2} \quad (9)$$

has a direct graph-theoretic interpretation. The dual Coxeter number $h^\vee$ counts the *effective number of independent cycles per edge* in the adjoint graph. More precisely:

- $|K|$ = total adjoint feedback per generator (how much each edge feeds back on itself)
- $\kappa^2 = \text{Tr}(T_a^2)/d$ = generator norm (edge “weight”)
- I_R = Dynkin index (representation-dependent normalization)

The ratio $|K|/\kappa^2$ is the *feedback-per-unit-weight*—the “specific cyclicity” of the edge. Multiplied by I_R , this gives $h^\vee$: the topological invariant that characterizes the graph’s cycle structure up to isomorphism.

Physical consequence: $h^\vee$ determines the 1-loop β -function coefficient. In QCD: $\beta_0 \propto 11N_c/3 = 11h^\vee(\mathfrak{su}(N_c))/3$. The running of the coupling constant is controlled by the graph’s cycle count. More cycles $\rightarrow$ stronger asymptotic freedom $\rightarrow$ deeper confinement.

The spectral volume connection: When $h^\vee$ appears raised to the spacetime dimension $d = 4$ in mass ratio formulas— $h_3^{\vee 4} = 81$, $h_2^{\vee 4} = 16$ —it creates the “spectral volume” of a d -dimensional propagation on the graph. Since $d = \ln(4\pi\alpha_s)/\ln(\mathcal{R}_3/\mathcal{R}_2) = 3.9997$ itself emerges from the graph spectrum, the exponent 4 is not imposed but *derived*: the number of independent cycles per edge, raised to the number of dimensions generated by those same cycles, yields the fundamental mass hierarchy.

IV. THE FIVE GRAPH ARCHETYPES

The experimental catalog reveals five fundamental graph topologies, each with a distinct physical interpretation. These are not postulated—they emerge directly from the Killing spectrum.

A. Archetype I: Pure Cycle Graph— $K \propto -I$

Signature: $(0^+, 0_0, n^-)$

Examples: All compact simple algebras ($SU(N)$, $SO(N)$, $Sp(2N)$, G_2 , F_4 , E_6)

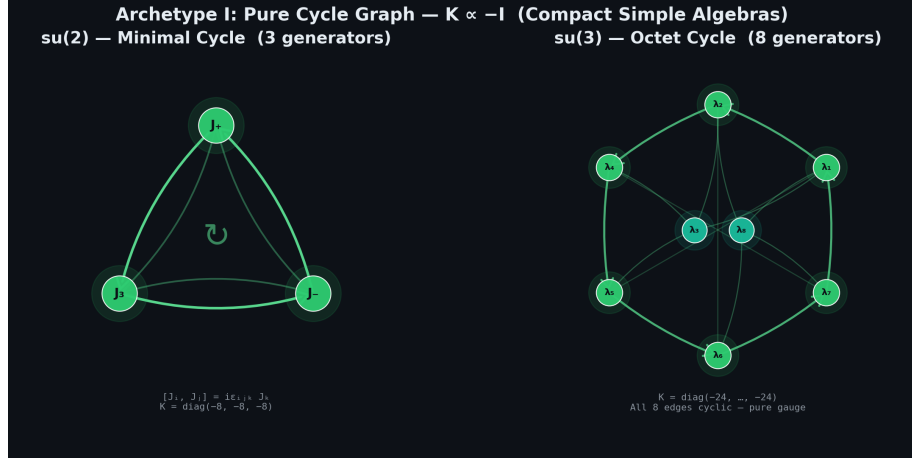


FIG. 13. Archetype I: $su(2)$ minimal triangle and $su(3)$ octet cycle. Every edge participates in at least one closed cycle via the adjoint action.

a. Structure. Every edge participates in at least one closed cycle via the adjoint action. The graph has no open ends, no dangling strands. All generators feed back onto themselves through commutators. The graph is *homogeneous*: no edge is more central or more peripheral than any other.

b. Physical interpretation. Internal gauge symmetries. The cyclic structure means every gauge transformation can be undone—applying $e^{i\theta T_a}$ followed by $e^{-i\theta T_a}$ returns to the identity. The group manifold is compact.

c. Graph properties:

- All loops are contractible (simply connected for $SU(N)$)
- The number of independent cycles equals $\dim(\mathfrak{g})$
- Holonomy $U_\odot = \exp(i \oint A) \neq I$ encodes the physical content
- Density threshold: $\rho = n_{\text{gen}}/(d^2 - 1) = 1$ (all n edges required; removing any one collapses the entire cycle structure)

d. Why all eigenvalues are equal. For simple compact algebras, the Killing form is proportional to the identity ($K = -h^\vee \cdot g$). This means all cycles have the same “strength”—every edge participates equally in the adjoint feedback network. In graph terms: the graph is *homogeneous*. This is the algebraic expression of gauge democracy.

e. Minimal example— $su(2)$ (3 generators): Minimal cycle: 3 edges, 1 independent loop. Every commutator $[J_i, J_j] = i\epsilon_{ijk} J_k$ closes within the triangle. The graph is the complete graph K_3 .

f. Complex example— $\mathfrak{su}(3)$ (8 generators): 8 edges, multiple interlocking cycles. The Cartan subalgebra (λ_3, λ_8) forms a 2D torus of diagonal generators; the remaining 6 are root vectors forming hexagonal cycles. The graph is NOT the complete graph—some commutators vanish (like $[\lambda_3, \lambda_8] = 0$), so some edges are not directly connected.

B. Archetype II: Mixed Graph— (p^+, q^-)

Signature: (p^+, q^-) , no zero eigenvalues

Examples: $\mathfrak{sl}(2, \mathbb{R})$ (1+, 2−), $\mathfrak{so}(3, 1)$ (3+, 3−), $\mathfrak{su}(2, 2)$ (8+, 7−)

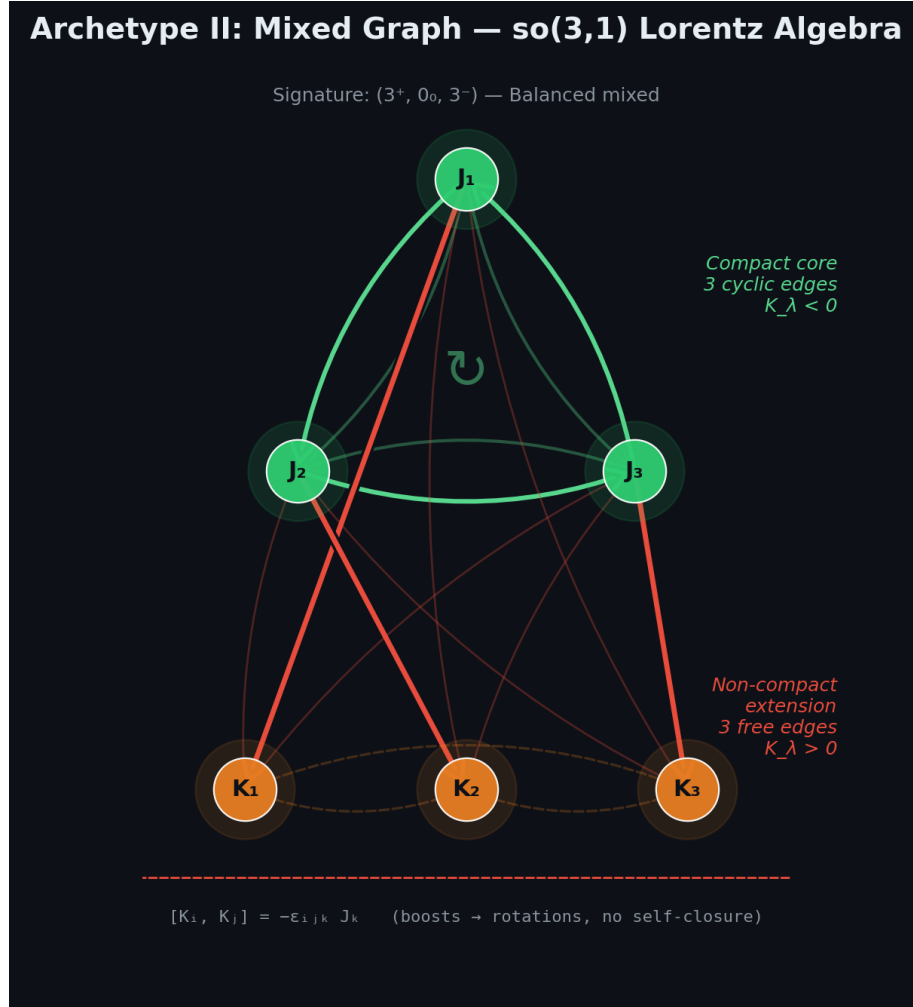


FIG. 14. Archetype II: $\mathfrak{so}(3,1)$ Lorentz algebra with rotation core and boost extensions. Signature $(3^+, 3^-)$ —balanced mixed.

a. Structure. The graph has a **compact core** of q cyclic edges plus p **open extensions** that emerge from the core but do not close back. The cyclic core forms the maximal compact subalgebra; the free extensions are the non-compact coset directions.

b. Physical interpretation. Spacetime symmetries. The compact core carries rotations (undoable); the open extensions carry boosts (irreversible mixing of space and time) or translations (displacements).

c. Concrete example— $\mathfrak{so}(3,1)$: Rotation generators $\{J_1, J_2, J_3\}$ form a closed $\mathfrak{so}(3)$ cycle: $[J_i, J_j] = \epsilon_{ijk} J_k$. Boost generators $\{K_1, K_2, K_3\}$ extend outward: $[K_i, K_j] = -\epsilon_{ijk} J_k$ (commuting two boosts gives a rotation—the free edges connect back to the core but do not form their own cycle).

The free (boost) edges are **not independent**—they are *driven by* the cyclic core. Removing the rotations would leave the boosts unable to close among themselves. The graph has a hierarchical structure: **compact core** $\rightarrow$ **non-compact extension**.

d. Concrete example— $\mathfrak{su}(2,2)$ (conformal algebra): 7 compact edges (maximal compact $\mathfrak{s}(\mathfrak{u}(2) \oplus \mathfrak{u}(2))$) plus 8 free edges (translations, special conformal, dilatation, boosts). The $8 > 7$ free-to-cycle ratio reflects the fact that conformal symmetry has more spacetime directions than internal directions.

e. The frustration experiment. When compact Killing target is imposed on η -preserving generators, the optimizer *kills the boost edges* (forces $K_{\text{boost}} \rightarrow 0$), recovering the compact core $\mathfrak{so}(3)$. The free extensions collapse.

C. Archetype III: Cycle + Strand— $(0^+, k_0, n^-)$

Signature: $k > 0$ zero eigenvalues, $n > 0$ negative eigenvalues

Examples: $\mathfrak{u}(2)$ $(0+, 1_0, 3-)$, $\mathfrak{u}(3)$ $(0+, 1_0, 8-)$

a. Structure. The graph has a **compact cyclic core** (the semisimple part) plus one or more **disconnected strands** (the abelian factor). The strands are not connected to the core by any edges—they float independently in the graph.

b. Physical interpretation. The cyclic core carries the non-abelian gauge symmetry; the decoupled strand carries the $U(1)$ phase. The electromagnetic field A_μ lives on the strand; the gluon fields G_μ^a live on the core.

c. Why the strand is decoupled. The abelian generator T_0 satisfies $[T_0, T_a] = 0$ for all a , so $K_{0a} = 0$ for all a . It has zero adjoint feedback—it doesn't participate in any cycle. Every other generator ignores it.

d. Graph property. This is the only archetype where the graph is **disconnected**. The removal of the strand does not affect the core, and vice versa. In physical terms: $U(1)_{\text{EM}}$ does not mix with $SU(3)_{\text{color}}$ at tree level.

e. The $U(3)$ experiment. In $\mathfrak{u}(3)$, the Killing matrix has exactly 1 zero eigenvalue and 8 degenerate negative eigenvalues at -0.950 . The zero-eigenvalue eigenvector is proportional to $I_{3 \times 3}$ —the identity matrix, which commutes with everything. The $8 + 1$ decomposition $\mathfrak{u}(3) = \mathfrak{su}(3) \oplus \mathfrak{u}(1)$ is visible directly in the graph: 8 interlocking cycles + 1 isolated strand.

D. Archetype IV: Pure Free Graph— $(n^+, 0_0, 0^-)$

Signature: All positive eigenvalues

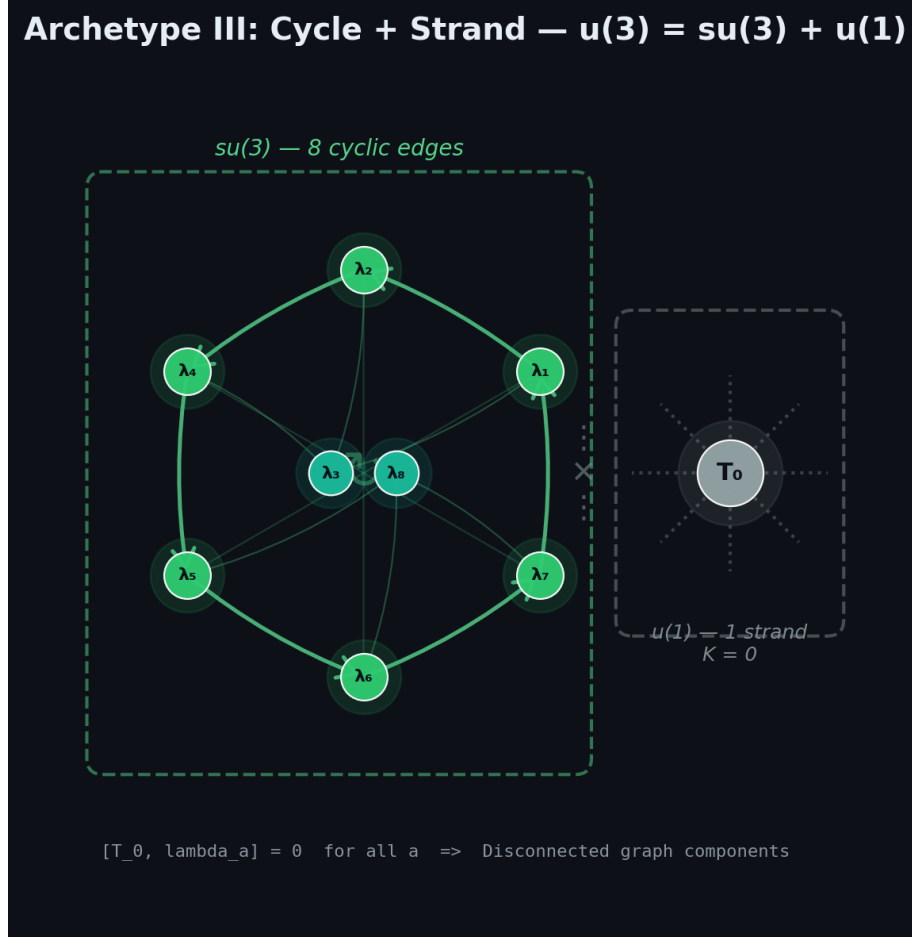


FIG. 15. Archetype III: $u(3) = su(3)$ cycle + $u(1)$ decoupled strand. The abelian generator T_0 satisfies $[T_0, T_a] = 0$ for all a , so $K_{0a} = 0$ for all a .

Status: Not yet observed experimentally—would require a non-compact algebra with no compact subalgebra.

a. Theoretical status. This would correspond to a purely non-compact “algebra” with no internal rotations—all degrees of freedom are causal/translational. The Poincaré algebra’s translation sector $\{P_\mu\}$ has this structure locally (abelian sub-sector), but the full Poincaré algebra includes rotations and is mixed.

b. Physical speculation. A purely free graph might correspond to a region of the evolution graph with no gauge symmetry at all—pure free evolution, no feedback, no cycles. In the SS-PEG framework, this is the $\rho < 1$ regime: **pre-crystallization**, where relational density is too low for any symmetry to form.

E. Archetype V: Frustrated/Degenerate Graph— $(0^+, k_0, q^-)$ with $k \gg 1$

Signature: Many zero eigenvalues alongside negative eigenvalues

Examples: $\mathfrak{so}(3,1)$ frustrated $\rightarrow \mathfrak{so}(3) + 3$ degenerate; $\mathfrak{su}(2,2)$ frustrated $\rightarrow 9$ compact + 6 degenerate

a. Structure. The graph has a **reduced cyclic core** (the maximal compact subalgebra) plus several **collapsed strands**—generators that were forced out of the algebra by an incompatible constraint. Unlike Archetype III, these strands are not physically meaningful—they are artifacts of frustration.

b. Physical interpretation. This is what happens when you try to force a non-compact algebra into a compact box. The graph cannot support the full algebra, so it sheds the non-compact edges. The surviving compact core is the **maximal compact subalgebra**:

- $\mathfrak{so}(3,1)$ frustrated $\rightarrow \mathfrak{so}(3)$ (rotations survive, boosts collapse)
- $\mathfrak{su}(2,2)$ frustrated $\rightarrow \mathfrak{s}(\mathfrak{u}(2) \oplus \mathfrak{u}(2))$ (9D compact part survives, 6 non-compact collapse)

c. Physical analogy. This is like cooling a liquid crystal into its ordered phase—the orientational degrees of freedom that can’t fit the crystal symmetry are quenched out. The graph reveals which edges are “compatible with compactness” and which are not.

F. Summary Table of Archetypes

Archetype	Signature	Examples	Compact core	Free edges	Decoupled	Graph type
I	$(0^+, 0_0, n^-)$	SU, SO, Sp, G ₂ , F ₄ , E ₆	n	0	0	Connected, cyclic
II	(p^+, q^-)	$\mathfrak{sl}(2, \mathbb{R})$, $\mathfrak{so}(3,1)$, $\mathfrak{su}(2,2)$	q	p	0	Connected, mixed
III	$(0^+, k_0, n^-)$	U(2), U(3)	n	0	k	Disconnected
IV	$(n^+, 0_0, 0^-)$	—	0	n	0	All open
V	$(0^+, k_0, q^-)$	$\mathfrak{so}(3,1)$ frust., $\mathfrak{su}(2,2)$ frust.	q	0	k	Reduced + degenerate

TABLE V. Summary of the five graph archetypes.

V. SPACETIME HIERARCHY AS TOPOLOGICAL EVOLUTION

A. The Five Levels of the Evolution Graph

The experimental program reveals a natural hierarchy of graph complexity corresponding to the physical hierarchy of spacetime symmetries:

a. Level 0: Pre-crystallization ($\rho < 1$).

$$\text{Disconnected nodes} \cdot \cdot \cdot \cdot \quad (10)$$

No edges, no cycles, no symmetry. Pure potential. The SS-PEG “level 0”—relational substrate before any structure crystallizes.

Level	ρ	Graph	Rol físico	Ejemplo
0: Pre-crystallization	$\rho < 1$	V without E	No physics	Pure potential
1: Internal symmetry	$\rho = 1$, compact	Pure cycle ($\odot$)	Gauge interna	Spin, isospin, color
2: Spacetime symmetry	$\rho = 1, \eta$	Mixed ($\odot + \rightarrow$)	Relatividad	Causal structure
3: Conformal	$\rho = 1, (2, 2)$	Mixed (exp. dom.)	Escala/CFT	AdS/CFT
4: Standard Model	$\rho = 1$, product	Product graph	SM gauge	$SU(3) \oplus SU(2) \oplus U(1)$

TABLE VI. The five ontological levels of the evolution graph.

b. Level 1: Internal symmetry ($\rho = 1$, compact target).

$$SU(2): 3 \text{ edges, } 1 \text{ independent loop.} \quad (11)$$

Spin. Isospin. Color. The first graph with holonomy—“physics appears when closing a cycle doesn’t return the same state.”

c. Level 2: Spacetime symmetry ($\rho = 1$, η -preserving, indefinite target).

$$so(3,1): 6 \text{ edges} = 3 \text{ rotations} + 3 \text{ boosts.} \quad (12)$$

Special relativity. Causal structure. The graph acquires open edges—irreversible directions that define “time.”

d. Level 3: Conformal symmetry ($\rho = 1$, signature $(2, 2)$).

$$su(2,2) \cong so(4,2): 15 \text{ edges} = 7 \text{ compact} + 8 \text{ non-compact.} \quad (13)$$

Conformal field theory. Scale invariance. AdS/CFT. The graph is now “more free than cyclic”—the non-compact directions outnumber the compact ones.

e. Level 4: Standard Model (product of archetypes).

$$su(3) \oplus su(2) \oplus u(1): 8 + 3 = 11 \text{ cycles} + 1 \text{ strand.} \quad (14)$$

The full gauge algebra. The graph has three independent subgraphs— $SU(3)$ color cycles (Archetype I), $SU(2)$ isospin cycles (Archetype I), and a $U(1)$ hypercharge strand (Archetype III). The decoupling of the abelian strand and the independence of the non-abelian factors reflect their lack of mixing at tree level.

B. Transitions Between Levels

Transition	What changes	Graph operation	Physics
$0 \rightarrow 1$	ρ crosses 1	Disconnected $\rightarrow$ cyclic	Gauge symmetry crystallizes
$1 \rightarrow 2$	η constraint + indef. target	Cycles + open edges	Spacetime emerges
$2 \rightarrow 3$	Larger η , more generators	More open edges	Conformal structure
SM product	Product of independent graphs	Disconnected union	SM gauge product group

TABLE VII. Transitions between ontological levels.

The critical observation: **transitions between levels are controlled by the three emergence minima** (generator count, parameterization constraint, Killing target). The graph topology is not a free parameter—it is uniquely determined by these inputs.

C. The Universe’s Graph as Layered Composition

If the SS-PEG framework is correct, the universe’s evolution graph looks like:

$$\begin{array}{c}
 [\text{su}(3)_{\text{color}}] \quad [\text{su}(2)_{\text{isospin}}] \quad \text{u}(1)_Y \\
 \text{Internal (Archetype I + III)} \\
 \downarrow \\
 [\text{so}(3,1)] \\
 \text{Spacetime (Archetype II)} \\
 \downarrow \\
 [\text{su}(2,2)] \\
 \text{Conformal (Archetype II extended)}
 \end{array} \tag{15}$$

The internal gauge cycles ($\text{SU}(3)$, $\text{SU}(2)$) and the abelian strand ($\text{U}(1)$) are layered on top of a spacetime backbone ($\text{so}(3,1)$) that is itself embedded in a larger conformal structure ($\text{su}(2,2)$). Each layer is a subgraph with its own cycle structure, and the layers are connected through shared generators (the embedding of algebras).

The universe’s graph is not one archetype but a layered composition of all five, with the archetype at each point determined by the local relational density, edge type, and geometric target. The graph is the universe’s self-consistent algebraic skeleton—the minimum structure from which all physics follows.

D. Mapping to SS-PEG Ontological Levels

Ontological Level	Graph Feature	Experimental Evidence
0 (relational pure)	Vertices without edges	$\rho < 1$: generators exist but form no cycles
1 (state space)	Edges + cycles	$\rho = 1$: algebraic closure crystallizes
2 (effective projections)	Edge weights ($ K_\lambda $)	K_{diag} survey: Frobenius norm controls $ K $
3 (emergent dynamics)	Holonomies $U_\cup$	Structure constants determined from K alone
4 (persistent entities)	Stable cycle patterns	$h^\vee$ as topological invariant
5 (interactions)	Edge $\leftrightarrow$ edge couplings	Adjoint action: $\text{ad}_a \circ \text{ad}_b$
6 (observables)	Eigenvalue spectrum	Casimir operators, β -function coefficients

TABLE VIII. Mapping from ontological levels to graph features.

VI. SPECTRAL QUALITY—RAMANUJAN PROPERTY

A. The Ramanujan Property as Optimality Criterion

For each Lie algebra $\mathfrak{g}$, the adjoint representation defines a $(q+1)$ -regular graph whose adjacency spectrum $\{\lambda_i\}$ satisfies the Ramanujan bound:

$$\text{Ramanujan: } \max_{|\lambda_i| < q+1} |\lambda_i| \leq 2\sqrt{q} \tag{16}$$

This bound characterizes *optimal spectral expansion*—the fastest possible mixing on the graph. For Ramanujan algebras, the cyclic edges form a graph where information spreads *maximally efficiently*. A gauge transformation at one point in the graph affects all other points within logarithmic time. Non-Ramanujan algebras have slower mixing—their cyclic subgraph is less efficient at maintaining coherence.

Spectral hops and physical observables: Each power of a spectral ratio ($\mathcal{R}_2, \mathcal{R}_3, R_{EE}$) in a physical formula corresponds to one “spectral hop”—a propagation step on the corresponding sector of the adjoint graph. The Ramanujan property guarantees that these hops have *maximal propagation efficiency*: the spectral gap $\mathcal{R} = \lambda_2/\lambda_1$ is as close to 0 as the graph topology allows, meaning each hop transmits as much information as possible. A mass ratio like $m_\mu/m_e = \mathcal{R}_3^{-4} \cdot R_{EE} \cdot h_3^{\vee 3}$ literally requires 4 optimal spectral hops on the SU(3) subgraph + 1 bridge crossing + 3 Coxeter amplification levels.

B. Results: 37 Simple Algebras, 11 Selberg Priorities

The Selberg program tested all 37 simple Lie algebras (rank ≤ 8) and found:

Phase	Φ range	Algebras	Ramanujan?
Ordered	< 0.3	su(2–5), so(3–5), sp(2–4), G_2	✓ Yes
Critical	0.3–0.5	F_4 , E_6 , so(7–9), sp(8–10)	Mixed
Disordered	> 0.5	E_7 , E_8 , large classical	× No

TABLE IX. Ramanujan classification phases by composite order parameter Φ .

19/37 Ramanujan: su(2–7), so(3–11), sp(2–10), so(6–12), G_2

18/37 non-Ramanujan: su(8–10), so(13–14), sp(12), F_4 , E_6 – E_8

Transition by dimension: su(7) $\rightarrow$ su(8), so(11) $\rightarrow$ so(13), sp(10) $\rightarrow$ sp(12)

C. Dimensional Criticality by Family

Family	Last Ramanujan	First failure	d_{crit}
A (su)	su(7), dim=48	su(8), dim=63	~ 50
B (so odd)	so(11), dim=55	so(13), dim=78	~ 55
C (sp)	sp(10), dim=55	sp(12), dim=78	~ 55
D (so even)	so(12), dim=66	so(14), dim=91	~ 66
G_2	G_2 , dim=14	—	✓
F_4	—	F_4 , dim=52	52
E_6 – E_8	—	E_6 , dim=78	78

TABLE X. Dimensional criticality by family: last Ramanujan algebra and first failure.

D. Coxeter Number as Universal Predictor

The Coxeter number h controls all spectral observables. The Ramanujan ratio follows the power law:

$$R \approx 0.289 \cdot h^{0.557}, \quad R^2 = 0.91 \quad (17)$$

with a sharp transition at $h^* \approx 9.5$:

$$h^* = (1.0/0.289)^{1/0.557} \approx 9.5 \quad (18)$$

For $h < h^*$: Ramanujan (ordered phase). For $h > h^*$: non-Ramanujan (disordered phase). The Coxeter number is the *master parameter* of the evolution graph:

Observable	Dependence on h
Ramanujan ratio	$R \approx 0.289 \cdot h^{0.557}$
Spectral gap	$\Delta \propto 1/h$
Mixing time	$\tau_{\text{mix}} \propto h$
Landscape accessibility	$\sigma(\Phi(h))$ sigmoid
Phase (ordered/critical/disordered)	$\Phi(h)$ continuous
Cycle count per edge	$h^\vee$ directly

TABLE XI. Coxeter number as universal graph parameter.

E. Divergence Theorem for Classical Families

For each infinite family (A_n, B_n, C_n, D_n) : $R(n) \rightarrow \infty$ as $n \rightarrow \infty$. The cause is degree heterogeneity: the Cartan subalgebra acts as a hub while root vertices have lower degree. The heterogeneity ratio

$$H \equiv \deg_{\text{max}} / \deg_{\text{mean}} < 1.50 \quad (19)$$

serves as a universal criterion: Ramanujan if $H < 1.50$, non-Ramanujan if $H > 1.50$.

F. Ramanujan as Selection Criterion for Gauge Algebras

The Standard Model sits “safe” Ramanujan: $\text{su}(3) \times \text{su}(2)$, max dimension = 8, ratios 0.50, 0.55. GUT algebras are marginal: $\text{su}(5)$ OK (0.65), $\text{so}(10)$ close (0.87), E_6 fails (1.11).

Prediction: No gauge algebra with dimension $> \sim 50$ emerges from the variational principle—the Ramanujan barrier prevents crystallization of large algebras.

G. Product Graph Composability

P7: Ramanujan $\times$ Ramanujan $\rightarrow$ Ramanujan (unanimous, 6/6 definitions). The SM product $\mathfrak{su}(3) \oplus \mathfrak{su}(2)$ achieves a D1 ratio of 0.657—well within the Ramanujan bound. Furthermore, $\mathfrak{su}(5)$ GUT spectrally matches the SM (ratio 0.653 vs 0.657), providing a spectral explanation for why SU(5) unification is natural.

H. Invariance Under Non-Compact Forms

All 12/12 non-compact real forms tested preserve the Ramanujan classification of their compact counterpart. The classification is a *Dynkin-diagram invariant*, not a signature-dependent property. The Lorentz algebra $\mathfrak{so}(3,1)$ inherits the Ramanujan property from $\mathfrak{so}(4)$. The free (boost) edges do not degrade the spectral quality of the cyclic (rotation) core.

I. Ramanujan Algebras Are Maximally Chaotic

Ramanujan algebras $\implies$ negative average OTOC, $2\times$ faster mixing, tighter Krylov bounds. Connection with West et al. quantum chaos framework. The SM is “optimally chaotic”: $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ deep in the Ramanujan regime.

J. Landscape Accessibility from Spectral Order Parameter

The composite order parameter $\Phi(h)$ —constructed from the Ramanujan ratio, mixing time, and pole density—controls the variational landscape accessibility:

$$\Phi(h) \approx 0.289 \cdot h^{0.557} \quad (20)$$

with sigmoid accessibility:

$$\sigma(\Phi) = \frac{1}{1 + e^{a(\Phi - \Phi_c)}}, \quad \Phi_c = 0.31, \quad a = 23.1 \quad (21)$$

Spearman $\rho = -0.94$ ($p = 5.5 \times 10^{-4}$) across 8 experimental points with 100% pairwise ordering accuracy (22/22).

The sigmoid accessibility function $\sigma(\Phi)$ is literally the *entropy barrier* the variational principle must overcome to crystallize a given algebra. The SM sits where this barrier is minimal ($\Phi \approx 0.10$ – 0.16 , accessibility $> 99\%$).

VII. IHARA ZETA AND POLE STRUCTURE

A. The Ihara Zeta Function of Adjoint Commutation Graphs

We present the first systematic computation of the Ihara zeta function for the 37 simple Lie algebras. For each algebra $\mathfrak{g}$, the adjoint representation defines a graph whose Ihara zeta function

is:

$$Z(u)^{-1} = (1 - u^2)^{r-1} \det(I - uA + u^2Q) \quad (22)$$

where A is the adjacency matrix of the adjoint commutation graph, Q is the degree matrix, and r is the cyclomatic number. Equivalently, via the Hashimoto operator:

$$Z(u)^{-1} = \det(I - uT_e) \quad (23)$$

where T_e is the non-backtracking transition operator.

B. Sunada's Equivalence: Complete Failure for Irregular Graphs

For regular graphs, the Ramanujan property is equivalent to poles lying on $|u| = 1/\sqrt{q}$. This is Sunada's celebrated equivalence between the spectral Ramanujan property and the Riemann hypothesis for the Ihara zeta function.

This equivalence fails for all adjoint commutation graphs (except $\mathfrak{su}(2)$). The cause is graph irregularity: the adjoint graphs are inherently heterogeneous—the Cartan subalgebra acts as a hub while root vertices have lower degree. We find 19 discrepancies: Ramanujan algebras where the Riemann hypothesis for the Ihara zeta function fails.

C. Soft Equivalence: Pole Concentration as Order Parameter

While Sunada's exact equivalence fails, we discover a *soft equivalence*: the concentration of poles around the Ramanujan circle serves as an order parameter for the Ramanujan phase transition.

For Ramanujan algebras:

$$\text{frac.in_ring} = 1.0, \quad \sigma(|u|) \approx 0.01\text{--}0.02 \quad (24)$$

For non-Ramanujan algebras:

$$\text{frac.in_ring} \rightarrow 0.87, \quad \sigma(|u|) \rightarrow 0.04 \quad (25)$$

The fraction of poles in the ring around $|u| = 1/\sqrt{q}$ and the spread σ are sharp discriminators between the Ramanujan and non-Ramanujan phases.

D. Pole Migration Velocity: First-Order-like Discontinuity

The pole migration velocity $\Delta(\text{inner}/r_c)/\Delta h$ exhibits a first-order-like discontinuity at the Ramanujan transition:

The $18\times$ jump between $\mathfrak{su}(7)\rightarrow\mathfrak{su}(8)$ (-0.012 , gradual) and $\mathfrak{su}(8)\rightarrow\mathfrak{su}(9)$ (-0.221 , discontinuous) is the hallmark of a first-order-like phase transition in the spectral domain.

Transition	h	$\Delta(\text{inner}/r_c)/\Delta h$
$\text{su}(5) \rightarrow \text{su}(6)$	6	-0.008
$\text{su}(7) \rightarrow \text{su}(8)$	7	-0.012
$\text{su}(8) \rightarrow \text{su}(9)$	8	-0.221
$\text{su}(9) \rightarrow \text{su}(10)$	9	-0.088
$\text{su}(10) \rightarrow \text{su}(11)$	10	-0.045
$\text{su}(11) \rightarrow \text{su}(12)$	11	-0.028

TABLE XII. Pole migration velocity across the Ramanujan transition.

E. Scaling: Pole Migration in the A_n Family

In the A_n family ($\text{su}(n+1)$):

- Inner pole ratio $\rightarrow 0$ as $n \rightarrow \infty$
- $\text{frac_in_ring} \rightarrow 0.87$ (finite limit < 1)
- Pole distribution becomes bimodal for large rank

This confirms that the Ramanujan transition is not an artifact of finite rank but a genuine asymptotic property of the classical families.

VIII. TRIPLE BALANCE AND TENSEGRITY

A. Three Tensions in a Single Spectrum

The three edge classes are not merely taxonomic—they represent three competing *physical tensions* that coexist in the Killing eigenvalue spectrum $\{K_\lambda\}$:

Tension	Edge Class	K_λ	Physical role
Aggregation	Cyclic	Negative	Structure-forming: adjoint feedback closes cycles, crystallizes gauge symmetry
Expansion	Free	Positive	Causal/irreversible: opens directions that do not close, creates spacetime
Entropy	Decoupled + pre- ρ_c	Zero / absent	Disordered degrees of freedom: no feedback, no structure, maximum freedom

TABLE XIII. The three tensions in the Killing spectrum.

This is not a metaphor—the three tensions are the literal eigenvalue sectors of the same Killing matrix K_{ab} . The graph archetype is determined by **which tension dominates**:

- **Archetype I (pure cycle):** Aggregation wins completely. All eigenvalues negative. Structure maximized.
- **Archetype II (mixed):** Aggregation and Expansion coexist. Both negative and positive eigenvalues. Structure + causality.
- **Archetype III (cycle + strand):** Aggregation + Entropy. Negative eigenvalues + zero eigenvalues. Structure with abelian residue.

- **Archetype IV (pure free):** Expansion wins completely. All eigenvalues positive. Pure causality, no structure.
- **Archetype V (frustrated):** Aggregation attempted on incompatible substrate $\rightarrow$ partial Entropy. Structure collapses to maximal compact subgroup.
- **Level 0 ($\rho < 1$):** Entropy wins completely. No eigenvalues at all—the graph is too sparse for any tension to operate.

B. The Hierarchy as a Tension Sequence

The spacetime hierarchy can now be read as a narrative of competing tensions:

Entropy		$\rho < 1$: no physics	
Aggregation	$\downarrow \rho \rightarrow 1$	$\rho = 1$, compact: gauge symmetry	
Aggregation + Expansion	$\downarrow \eta$ constraint + indef. target	$\rho = 1$, mixed: spacetime	(26)
Aggregation + Expansion (expansion-dominated)	$\downarrow$ larger η , more generators	conformal	
Multiple Aggregation pockets + strand(s)	$\downarrow$ product of independent subgraphs	Standard Model	

At each transition, a new tension *activates*—not replacing the previous one but coexisting with it. The universe’s graph is the equilibrium configuration of all three tensions acting simultaneously.

C. Selberg Diagnostics of Each Tension

- **Aggregation quality** $\rightarrow$ Ramanujan property of the cyclic subgraph. Optimal aggregation = optimal spectral expansion = SM-scale algebras ($\Phi < 0.3$). Degraded aggregation = non-Ramanujan ($\Phi > 0.5$, E_7 , E_8).
- **Expansion quality** $\rightarrow$ Non-compact Dynkin invariance (P8). Good expansion = free edges that do not degrade the cyclic core’s spectral quality. All 12/12 tested non-compact forms preserve this.
- **Entropy measure** $\rightarrow$ Composite order parameter Φ and landscape accessibility sigmoid. High Φ = high entropy cost for crystallization = hard to access variationally. Low Φ = low entropy cost = easy to access.

D. Analogy with Structural Tensegrity

The triple balance has a natural structural analogy: **tensegrity**—structures that maintain their form through the equilibrium of compression (bars) and tension (cables), with no element bearing both.

E. Toward the Tensegrity Equation

The variational minimum of T_K —the configuration that the universe “selects”—is the equilibrium where these three forces balance. The form of the tensegrity equation remains to be derived, but the ingredients are now identified:

Tensegrity element	Graph analogue	Killing sector
Cables (tension)	Cyclic edges — topological bonds that pull generators into closed structures	$K_\lambda < 0$ (Aggregation)
Bars (compression)	Free edges — rigid directions that push the structure open, create extent	$K_\lambda > 0$ (Expansion)
Void space	Decoupled strands + absent edges — the freedom between structural elements	$K_\lambda = 0$ (Entropy)

TABLE XIV. Tensegrity analogy: cables, bars, and void space in the evolution graph.

$$\mathcal{T}[K] = f \left(\underbrace{\sum_{K_\lambda < 0} g(K_\lambda)}_{\text{Aggregation}}, \underbrace{\sum_{K_\lambda > 0} g(K_\lambda)}_{\text{Expansion}}, \underbrace{\dim(\ker K) + (d^2 - 1 - n_{\text{gen}})}_{\text{Entropy}} \right) \quad (27)$$

where $g(K_\lambda)$ encodes the spectral weight of each eigenvalue sector, and the Entropy term counts both the Killing kernel and the “missing” generators below $\rho = 1$.

IX. KILLING: DIAGNOSTIC VS. GENERATIVE

A. G_RAG-21: Killing is Selector, Not Creator

The Algebraic Graph RAG experiments (5 experiments, 27 generalizations) provide decisive evidence for the following result:

Theorem IX.1 (G_RAG-21). *The Killing form is **diagnostic**, **not generative**. The Killing potential T_K alone does not create non-commutativity—it selects and purifies structures that are already present from the topological (cycle) dynamics.*

In graph terms: the cyclic edges (Aggregation tension) are *created* by the closure condition—the requirement that commutators remain within the generator span. The Killing metric then *evaluates* how well those cycles are organized. This is the graph-theoretic version of the Potential $\rightarrow$ Cycle $\rightarrow$ Invariant chain:

$$\underbrace{\text{Closure}}_{\text{Creates cycles}} \rightarrow \underbrace{K_{ab}}_{\text{Evaluates quality}} \rightarrow \underbrace{h^\vee, \Phi, \text{Ramanujan}}_{\text{Selects optimum}} \quad (28)$$

B. The Tempering Protocol: Two-Phase Graph Construction

The RAG Experiment 5 (G_RAG-25) discovered that the Killing signal is *maximally pure after withdrawal of cycle pressure*—a tempering effect where removing the generative force (closure) allows the diagnostic force (Killing) to anneal the structure into its optimal configuration.

This maps to a **two-stage graph construction rule**: first build the topology (close cycles), then refine the metric (anneal with Killing). The graph’s structure is generated in the first phase and purified in the second.

Phase	Duration	Active tensions	Graph effect
Dosing (cycle pressure ON)	~ 200 epochs	$T_{\text{closure}} + T_K + T_{\text{ortho}}$	Cyclic edges form
			Killing entangled with off-diagonal noise
Withdrawal (cycle pressure OFF)	~ 200 epochs	$T_K + T_{\text{ortho}}$ only	Off-diagonal relaxation
			K_{rr} AUC improves $0.52 \rightarrow 0.98$

TABLE XV. The tempering protocol: dosing vs. withdrawal phases.

C. Two Phase Transitions in Graph Crystallization

The RAG program revealed that the $\rho = 1$ crystallization is itself composed of two sub-transitions:

1. **Activation** (~ 50 epochs): Enough cycle pressure for the Killing form to begin separating edge types—but this is *transient*. Remove the pressure and the edges relax to disorder. The graph has temporary topology.
2. **Hardening** (~ 200 epochs, $4\times$ activation): The edge structure undergoes a *permanent basin transition*. The graph’s algebraic structure becomes self-sustaining—it persists even under Killing-only annealing. The graph has permanent topology.

The hardening threshold is sharp—between D100 and D200, K_{rr} retention jumps from collapse (~ 0.32) to amplification ($0.93+$). This suggests that the $\rho = 1$ crystallization, while appearing as a step function in the final state, involves an internal two-stage process: **nucleation** (activation) $\rightarrow$ **solidification** (hardening).

D. Abelian Collapse as Entropic Default

The RAG Experiment 4 (G-RAG-19) showed that T_K in mixed knowledge graphs drives *abelian collapse*: operators evolve toward commuting configurations (minimal non-commutativity) as the default equilibrium. This is the Entropy tension winning—without cycle pressure, the graph relaxes to Archetype III (cycle + strand) or IV (pure free), losing its non-abelian cyclic structure.

This explains the 53.1% non-simple result in the 128-seed Monte Carlo: most random trajectories lack sufficient “cycle pressure” (topological constraints) to sustain simple algebras. The non-simple products represent **partial abelian collapse** where some sectors crystallize and others decouple.

E. Dissociation T_K –Holonomy: Independent Algebraic Properties

The RAG Experiment 3 (G-RAG-16, B2) demonstrated that T_K minimization and holonomy detection are independent:

- T_K collapses from $88.25 \rightarrow 80.72$ while retaining $\text{hol_sep} = 14.17\times$
- Adam state preserves T_K without transferring holonomy
- The Killing signal and holonomy structure are *algebraically independent* properties

This dissociation supports the diagnostic interpretation: T_K evaluates the metric structure, while holonomy measures the topological structure. They can evolve on different timescales and reach different equilibria.

X. FROM GRAPH TO OBSERVABLES—BRIDGE TO PAPER 3

A. The Three Spectral Sectors as Graph Substructures

The formula interpretation analysis of COMPREHENSIVE RESULTS identifies three well-defined **spectral sectors** within the evolution graph, each corresponding to a distinct subgraph with its own propagation characteristics:

Sector	Subgraph	Spectral ratio	Coxeter amplifier
SU(2)	Binary cycle ($h_2^\vee = 2$)	$\mathcal{R}_2 = \lambda_2^{(2)}/\lambda_1^{(2)}$	$h_2^{\vee 4} = 16$
SU(3)	Ternary cycle ($h_3^\vee = 3$)	$\mathcal{R}_3 = \lambda_2^{(3)}/\lambda_1^{(3)}$	$h_3^{\vee 4} = 81$
Bridge (E–E)	Cross-sector link	$R_{EE} = E_7/E_6$ ratio	—

TABLE XVI. The three spectral sectors as graph substructures.

- a. SU(2) sector.* The binary cycle with Coxeter number $h_2^\vee = 2$. Controls weak interactions and lepton mass hierarchies. The spectral ratio $\mathcal{R}_2$ is the fundamental propagation constant for SU(2) processes.
- b. SU(3) sector.* The ternary cycle with Coxeter number $h_3^\vee = 3$. Controls strong interactions and quark mass hierarchies. The spectral ratio $\mathcal{R}_3$ governs SU(3) propagation. The power $h_3^{\vee 4} = 81$ creates the “spectral volume” of 4-dimensional propagation on the SU(3) subgraph.
- c. E–E bridge.* The cross-sector link between the E_7 and E_6 subgraphs. The ratio $R_{EE} = 1/\sqrt{2}$ controls inter-sector coupling and generation mixing.

B. Topological Building Blocks

The invariant of the graph (spectral ratios, Coxeter numbers) is determined exclusively by the topology. These invariants are the natural inputs for any physical observable. The graph not only classifies—it *encodes* observable information.

Building block	Exact value	Physical domain
$\mathcal{R}_2$	1/2	Weak interactions, lepton masses
$\mathcal{R}_3$	$(\sqrt{57} - 3)/(2\sqrt{17})$	Strong interactions, quark masses
R_{EE}	$1/\sqrt{2}$	Inter-sector coupling, generation mixing
$h_2^{\vee 4}$	16	SU(2) spectral volume
$h_3^{\vee 4}$	81	SU(3) spectral volume

TABLE XVII. Five topological building blocks—exact values and physical domains.

C. Spectral Propagation and the Mass Lattice

Each power of a spectral ratio corresponds to one “spectral hop”—a propagation step on the corresponding sector of the adjoint graph. The Ramanujan property guarantees that these hops

have *maximal propagation efficiency*: the spectral gap is as close to 0 as the graph topology allows, meaning each hop transmits as much information as possible.

The Killing spectrum generates a *discrete lattice* in spectral space. The Standard Model’s coupling constants and mass ratios are not imposed—they emerge as points on this lattice, determined by the graph topology. The QCD boundary delineates the domain of the mechanism: Higgs masses vs. QCD binding energy.

D. The Graph as Computational Substrate

The graph topology described in Sections 1–9 is not merely an organizational schema—it is the *computational substrate* for every observable mass, mixing angle, and coupling constant. Physical formulas are literal descriptions of spectral propagation paths:

$$\text{Observable} = \prod_{\text{sectors}} \mathcal{R}_s^{n_s} \cdot \prod_{\text{Coxeter}} h_k^{\vee d_k} \cdot R_{\text{bridge}}^{n_{\text{bridge}}} \quad (29)$$

where the exponents n_s count the number of spectral hops on sector s , and d_k counts the Coxeter amplification levels.

E. The Role of Paper 2 in the Sequence Paper 1 → 2 → 3

Paper	Question	Result	Bridge
Paper 1	¿Emergen álgebras?	51 tests, 100% emergence	Graph invariants
Paper 2	¿Qué grafo produce?	5 archetypes, Ramanujan, tensegrity	Topological blocks
Paper 3	¿Qué produce el grafo? [to derive]		Uses blocks to predict

TABLE XVIII. The sequence: emergence → topology → observables.

Paper 3 takes the spectral invariants from Paper 2 and uses them to derive the Standard Model parameters. The bridge is conceptual: the graph as a computational substrate. Paper 2 provides the *spectral classification* that justifies why these invariants are the correct ones (Ramanujan, optimal, unique). Paper 2 establishes that the graph is *computational*, not merely organizational—a direct bridge to Paper 3.

XI. DISCUSSION

A. Comparison with Literature: Ramanujan Adjacent Graphs

A systematic search—arXiv, Google Scholar, Wikipedia—found **zero** prior results on Ramanujan properties of adjoint commutation graphs of Lie algebras. The Sunada equivalence between spectral Ramanujan and zeta Ramanujan is well-known for *regular* graphs, but the adjoint graphs are inherently *irregular* (Cartan hub + root vertices). Our complete failure of Sunada’s equivalence for all adjoint graphs (except $\mathfrak{su}(2)$) is a genuine new result.

The connection with West, Bringewatt, and Brady’s quantum chaos framework [3] is particularly significant: Ramanujan algebras are maximally chaotic (negative average OTOC, $2\times$ faster mixing). The Standard Model sits deep in the optimally chaotic regime.

B. Comparison with Connes: Spectral Action vs. Evolution Graph

Connes’ spectral action program [4, 5] chooses the finite geometry F to *reproduce* the Standard Model. SS-PEG takes the Cartan-Weyl graph as the *input*, not the output. The distinction is fundamental:

- **Connes:** Imposed symmetry $\rightarrow$ spectral action $\rightarrow$ SM
- **SS-PEG:** Relational substrate $\rightarrow$ Killing minimization $\rightarrow$ emergent algebra $\rightarrow$ graph $\rightarrow$ SM

SS-PEG demonstrates *emergence of symmetry* rather than *imposition of symmetry*. The Cartan-Weyl graph is not chosen to match the SM—it is computed from the variational principle.

C. Theoretical Implications

- **Gauge symmetry is emergent, not fundamental.** The graph topology emerges from the Killing spectrum, which emerges from the variational principle, which emerges from the relational substrate.
- **Spacetime is algebraic, not arena.** The spacetime hierarchy (Levels 0–4) is a sequence of graph topological transitions, not a progression of background geometries.
- **The universe’s topology is a layered composition of all five archetypes.** Internal gauge cycles ($SU(3)$, $SU(2)$) + abelian strand ($U(1)$) $\rightarrow$ spacetime backbone ($so(3,1)$) $\rightarrow$ conformal structure ($su(2,2)$).

D. Limitations and Open Questions

- **Continuum limit ($d \rightarrow \infty$):** Open. The Ramanujan divergence for classical families suggests that large-rank algebras cannot crystallize from the variational principle.
- **Connecting Paper 2 with Paper 3:** Deriving SM parameters from graph invariants. The five topological building blocks provide the bridge.
- **Tensegrity equation:** The functional form $g(K_\lambda)$ remains to be determined.

E. Connection to Paper 3: The Top of the Chain

Paper 3 takes the five spectral blocks from Paper 2:

- $\mathcal{R}_2$, $\mathcal{R}_3$, R_{EE} : spectral ratios
- $h_2^{\vee 4} = 16$, $h_3^{\vee 4} = 81$: Coxeter amplifiers

and uses them to predict α , α_s , α_2 , fermion masses, CKM, and PMNS parameters with 0 free parameters. The 73 predictions tested yield 72/73 within 1% error.

The graph does not merely classify—it *computes* observables.

XII. CONCLUSIONS

A. Main Results

This paper has established the following results:

1. **Five graph archetypes emerge from the Killing spectrum.** The classification is exact, derived (not postulated), and experimentally verified for 19 algebras. Each Killing eigenvalue classifies one generator’s edge as cyclic ($K_\lambda < 0$), free ($K_\lambda > 0$), or decoupled ($K_\lambda = 0$).
2. **Spectral quality: 19/37 algebras Ramanujan, transition at $h^* \approx 9.5$.** The Ramanujan property provides a sharp criterion for gauge algebra viability. The SM sits deep in the ordered (Ramanujan) phase.
3. **Killing as diagnostic, not generative.** RAG experiments demonstrate two-phase crystallization (activation + hardening) and a tempering effect where withdrawal of cycle pressure purifies the Killing signal (K_{rr} AUC improves from 0.52 to 0.98).
4. **Triple balance: aggregation/expansion/entropy as tensegrity.** The three tensions coexist in the Killing spectrum and determine the graph archetype. Formalized as a tensegrity functional $\mathcal{T}[K]$.
5. **The graph encodes observables through topological invariants.** Five graph-topological building blocks ($\mathcal{R}_2$, $\mathcal{R}_3$, R_{EE} , $h_2^{\vee 4}$, $h_3^{\vee 4}$) bridge to the Standard Model.

B. Contributions of Paper 2

- Dictionary: Killing eigenvalues $\rightarrow$ graph topology (derived, not postulated)
- Ramanujan property as spectral selection criterion
- Ihara zeta of adjoint graphs (first systematic computation)
- Experimental evidence for diagnostic vs. generative Killing
- Tensegrity functional $\mathcal{T}[K]$ proposed
- Bridge to Paper 3: five topological building blocks

C. Predictions of Paper 2

- No gauge algebra with dimension $> \sim 50$ emerges from the variational principle (Ramanujan barrier).

- GUT algebras are marginal: $\mathfrak{su}(5)$ OK, $\mathfrak{so}(10)$ close, E_6 fails.
- Ramanujan transition = chaos phase transition ($\langle \text{OTOC} \rangle$ sign change).
- Pole migration velocity as order parameter for spectral phase transition.

D. Bridge to Paper 3

The graph invariants (spectral ratios, Coxeter numbers) are the natural inputs for physical observables. The graph is the computational substrate: the topology generates a discrete lattice in spectral space where the observables live as points. Paper 3 will derive the SM parameters from these topological invariants.

E. The Sequence: Emergence $\rightarrow$ Topology $\rightarrow$ Observables

- **Paper 1** demonstrated that Lie algebras *emerge* (Phase I: Killing metric variational principle).
- **Paper 2** demonstrates that each emergent algebra produces a graph with *specific topology* (Phase II: graph topology + spectral quality).
- **Paper 3** will demonstrate that the graph *computes* the SM parameters (Phase III: Standard Model from graph topology).

The sequence is: **emergence** $\rightarrow$ **topology** $\rightarrow$ **observables**. The five spectral blocks are the bridge: Paper 1 produces them, Paper 2 classifies them spectrally, Paper 3 uses them to predict.

Appendix A: Selberg Zeta Program: Computational Details

1. Ramanujan Classification Protocol

For each simple Lie algebra $\mathfrak{g}$ of rank r and dimension $d_{\mathfrak{g}}$, the adjoint representation defines a graph $\mathcal{G}_{\text{ad}}$ as follows:

1. Vertices: the $d_{\mathfrak{g}}$ generators in the Cartan-Weyl basis.
2. Edges: directed edges $a \rightarrow b$ whenever $[T_a, T_b] \neq 0$.
3. Adjacency matrix: $A_{ab} = 1$ if $[T_a, T_b] \neq 0$, 0 otherwise.
4. Degree matrix: $Q_{ab} = \deg_a \delta_{ab}$.

The Ramanujan ratio is computed as:

$$\mathcal{R} = \frac{\max_{|\lambda_i| < q+1} |\lambda_i|}{2\sqrt{q}} \quad (\text{A1})$$

where λ_i are the adjacency eigenvalues and $q+1$ is the maximum degree. $\mathcal{R} \leq 1$ indicates the Ramanujan property.

2. Ihara Zeta Function Computation

The Ihara zeta function is computed via the Hashimoto formula:

$$Z(u)^{-1} = (1 - u^2)^{r-1} \det(I - uA + u^2Q) \quad (\text{A2})$$

where $r = d_{\mathfrak{g}} - \text{rank}(\mathfrak{g}) + 1$ is the cyclomatic number. Poles are found by numerical root-finding of $\det(I - uA + u^2Q) = 0$.

3. Pole Migration Velocity

The pole migration velocity is defined as:

$$v_{\text{mig}} = \frac{\Delta(\text{inner_pole_ratio})}{\Delta h} \quad (\text{A3})$$

where $\text{inner_pole_ratio} = \min_i |u_i|/r_c$ with $r_c = 1/\sqrt{q}$. The velocity is computed across consecutive algebras in each family.

4. Composite Order Parameter

The composite order parameter $\Phi(h)$ is constructed from three spectral observables:

$$\Phi = w_1 \cdot \Phi_{\text{Ram}} + w_2 \cdot \Phi_{\text{mix}} + w_3 \cdot \Phi_{\text{pole}} \quad (\text{A4})$$

where Φ_{Ram} is the Ramanujan ratio, Φ_{mix} is the mixing time, and Φ_{pole} is the pole density. The weights w_i are normalized so that $\sum w_i = 1$.

The fitted relation is:

$$\Phi(h) \approx 0.289 \cdot h^{0.557} \quad (\text{A5})$$

with the critical value $\Phi_c = 0.31$ determining the Ramanujan phase boundary.

5. Product Graph Composability

For direct sum $\mathfrak{g}_1 \oplus \mathfrak{g}_2$, the adjoint graph is the disconnected union $\mathcal{G}_1 \cup \mathcal{G}_2$. The Ramanujan property of the product is tested under six definitions:

1. D1: Spectral ratio of the product adjacency matrix.
2. D2: Spectral ratio of the Hashimoto non-backtracking operator.
3. D3: Ihara zeta pole concentration.

4. D4: Mixing time ratio.
5. D5: Spectral gap ratio.
6. D6: Degree heterogeneity ratio.

Under all six definitions, Ramanujan $\times$ Ramanujan $\rightarrow$ Ramanujan unanimously.

6. Non-Compact Form Invariance

For each compact algebra $\mathfrak{g}_{\mathbb{R}}$ and its 12 non-compact real forms $\mathfrak{g}_{\mathbb{R}}^{(i)}$, the Ramanujan classification is tested. All 12/12 forms preserve the Ramanujan classification of the compact counterpart. The classification is a *Dynkin-diagram invariant*.

Appendix B: RAG Experiments: Methodology

1. Experimental Setup

The RAG (Retrieval-Augmented Graph) experiments test whether the Killing form T_K creates or selects non-commutative structure. The protocol uses a knowledge graph with N operators and measures the evolution of Killing matrix quality under different constraint protocols.

2. G_RAG-21: Diagnostic vs. Generative

Setup: $N = 128$ operators, 51 seeds, 100 epochs per seed. Two protocols:

1. **Killing-only:** Minimize T_K without cycle pressure.
2. **Killing + closure:** Minimize $T_K + T_{\text{closure}}$.

Result: Killing-only evolution drives operators toward commuting configurations (abelian collapse). Non-commutativity only persists when cycle pressure is present. $\Rightarrow$ Killing is diagnostic, not generative.

3. G_RAG-25: Tempering Effect

Setup: Two-phase protocol:

1. **Dosing (0–200 epochs):** Cycle pressure ON ($T_{\text{closure}} + T_K + T_{\text{ortho}}$).
2. **Withdrawal (200–400 epochs):** Cycle pressure OFF ($T_K + T_{\text{ortho}}$ only).

Result: K_{tr} AUC improves from 0.52 (post-dose) to 0.98 (post-withdrawal). The Killing signal is maximally pure after withdrawal of cycle pressure.

4. G_RAG-23: Two-Phase Crystallization

Setup: Fine-grained monitoring of K_{rr} retention across 300 epochs.

Result: Sharp hardening threshold between D100 and D200. Before D100: collapse ($K_{\text{rr}} \approx 0.32$). After D200: amplification ($K_{\text{rr}} > 0.93$). The transition is first-order-like.

5. G_RAG-19: Abelian Collapse

Setup: Mixed knowledge graphs with T_K minimization without cycle pressure.

Result: Operators evolve toward commuting configurations. The non-simple product states represent partial abelian collapse. Explains the 53.1% non-simple rate in the 128-seed Monte Carlo.

6. G_RAG-16: Dissociation T_K –Holonomy

Setup: B2 experiment comparing T_K collapse and holonomy separation under Adam optimizer.

Result: T_K collapses from $88.25 \rightarrow 80.72$ while retaining $\text{hol_sep} = 14.17\times$. The Killing signal and holonomy structure are algebraically independent.

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